

2010 高教社杯全国大学生数学建模竞赛

承 诺 书

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2. _____ 杨玉花 _____

3. _____ 袁定欢 _____

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摘 要

本文通过对储油罐中油位高度及变位参数之间的不同情形的储油量进行分析并建立相应的数学模型,在该过程中先利用投影法、截面法及微元法得出储油量与油位高度及变位参数的函数关系。再由 Matlab 编程可知各高度储油量的理论数据,最后分析误差及评价模型的合理性。

对于问题一的任一种情形,我们均建立笛卡尔坐标系,当储油罐无变位时,利用微元法得到体积关于 h 的公式,当储油罐发生变位时,根据储油罐中油量的多少分成三种情形,就每一类利用微元法得到体积关于 h 的公式。代人附件 1 实验数据中的高度得到储油罐中的理论油量 V 。根据理论油量及实际油量得出误差,判断误差所服从的分布,再利用相对误差进行误差分析并评价模型的合理性。由上述得到储油罐发生变位时体积关于 h 的公式我们给出了罐体变位后油位高度间隔为 1cm 的罐容表标定值(即进/出油量与罐内油位高度的表格)。

对于问题二中的储油罐,我们先将问题进行简化考虑,得出了储油罐水平卧放时油量与浮油子高度的函数关系;再考虑储油量与油位高度及变位参数(纵向倾斜角度 α 和横向偏转角度 β)的一般情况,在该过程中,我们进行近似处理,利用投影法和截面法得出了储油量关于油位高度及变位参数的函数关系;并在固定的横向偏转角度 β 条件下,就纵向倾斜角度 α 的变化进行分成三类讨论,这三类又可以分成八种情形,得到了每一种情形下实际储油罐罐内储油量与油位高度的函数关系。

在模型的改进中,我们就问题二储油量与油位高度及变位参数的一般情况进行仔细的考虑,将含油部分的体积分成四个部分,每一个部分将上述所提到的积分方法相结合,得到了各个部分的储油量与油位高度及变位参数的函数关系,从而可得总储油量与油位高度及变位参数的函数关系;并据此利用 Matlab 编程和实际测量的数据求得 α 和 β 值;与此同时我们可以得出在固定 α 、 β 值时各高度下的理论储油量;根据理论油量及实际油量得出误差,判断误差所服从的分布再利用相对误差进行误差分析并评价模型的合理性。由上述得到储油罐发生变位时体积关于 h 的公式我们给出了罐体变位后油位高度间隔为 10cm 的罐容表标定值。

【关键词】投影法 截面法 微元法 Matlab 编程

§ 1 问题重述

通常加油站都有若干个储存燃油的地下储油罐，并且一般都有与之配套的“油位计量管理系统”，采用流量计和油位计来测量进/出油量与罐内油位高度等数据，通过预先标定的罐容表（即罐内油位高度与储油量的对应关系）进行实时计算，以得到罐内油位高度和储油量的变化情况。

许多储油罐在使用一段时间后，由于地基变形等原因，使罐体的位置会发生纵向倾斜和横向偏转等变化（以下称为变位），从而导致罐容表发生改变。按照有关规定，需要定期对罐容表进行重新标定。

请你们用数学建模方法研究解决储油罐的变位识别与罐容表标定的问题。

（1）为了掌握罐体变位后对罐容表的影响，利用如图 4 的小椭圆型储油罐（两端平头的圆柱体），分别对罐体无变位和倾斜角为 $\alpha = 4.1^\circ$ 的纵向变位两种情况做了实验，实验数据如附件 1 所示。请建立数学模型研究罐体变位后对罐容表的影响，并给出罐体变位后油位高度间隔为 1cm 的罐容表标定值。

（2）对于图 1 所示的实际储油罐，试建立罐体变位后标定罐容表的数学模型，即罐内储油量与油位高度及变位参数（纵向倾斜角度 α 和横向偏转角度 β ）之间的一般关系。请利用罐体变位后在进/出油过程中的实际检测数据（附件 2），根据你们所建立的数学模型确定变位参数，并给出罐体变位后油位高度间隔为 10cm 的罐容表标定值。进一步利用附件 2 中的实际检测数据来分析检验你们模型的正确性与方法的可靠性。（以上涉及的图 1~4 均在附录中）

§ 2 模型的假设与符号的约定

§ 2.1 模型的假设与说明

- （1）在储油罐倾斜的情况下，忽略油浮子高度为 0 时油所占的体积；
- （2）在储油罐倾斜的情况下，假设当油浮子高度达到最大后不再进油；
- （3）油的挥发速度很慢，忽略因油的挥发而造成储油量的减少；
- （4）储油罐的材料为钢体，忽略因渗出油而造成储油量的减少；
- （5）储油罐管理妥当，不会因特殊情况而造成储油量的变化。

§ 2.2 符号的约定与说明

V 表示储油罐中油的体积；

L 表示储油罐圆柱体部分的长度；

a 表示任一椭圆截面的长半轴；

b 表示任一椭球截面的短半轴；
 a_1 表示油浮子在圆柱体高方向上投影至两端的较小值；
 h 表示油浮子到圆柱体高方向的距离；
 h_1 表示储油罐接地一端油面到地面得距离；
 α 表示纵向倾斜角度；
 β 表示横向倾斜角度；
 h_0 表示球冠高；
 b_0 表示球冠底半径；

§ 3 问题的分析

§ 3.1 问题一的分析

当储油罐无变位时，储油罐圆柱体的接地一端为原点，以圆柱体高方向为 z 轴，建立笛卡尔坐标系，利用微元法得到体积关于 h 的公式，代入附件 1 实验数据中的高度得到储油罐中的理论油量 V 。利用附件 1 实验数据中得到储油罐中的实际油量，根据理论油量及实际油量就可以得出误差，判断误差所服从的分布，利用相对误差进行误差分析。

当储油罐发生变位时，以储油罐圆柱体的接地一端为原点，圆柱体高方向为 z 轴，建立笛卡尔坐标系。根据储油罐中油量的多少分成三类，然后就每一类利用微元法得到体积关于 h 的公式，代入附件 1 实验数据中的高度得到储油罐中的理论油量 V 。利用附件 1 实验数据中得到储油罐中的实际油量，根据理论油量及实际油量就可以得出误差，判断误差所服从的分布，利用相对误差进行误差分析。

由上述得到储油罐发生变位时体积关于 h 的公式可以给出罐体变位后油位高度间隔为 1cm 的罐容表标定值（即进/出油量与罐内油位高度的表格）。

§ 3.2 问题二的分析

对于实际储油罐，我们首先将问题进行简化考虑，得出了当实际储油罐水平卧放时实际储油罐中油量与浮油子高度的函数关系；

然后我们先考虑实际储油罐罐内储油量与油位高度及变位参数（纵向倾斜角度 α 和横向偏转角度 β ）的一般情况，在该过程中，我们进行近似处理，利用投影法和截面法得出了储油量关于油位高度及变位参数的函数关系，再在固定的横向偏转角度 β 条件下，就纵向倾斜角度 α 的变化进行分类讨论，一共有三种情形，得到了每一种情形下实际储油罐罐内储油量与油位高度的函数关系。

最后我们先利用附件 2 中的少量实际数据得出了附件 2 所处状态下的纵向倾斜角度 α 和横向偏转角度 β ，再利用附件 2 中给定各高度进行代人，得到实际储

油罐理论的储油量，与实际储油量进行比较，求出误差及相对误差。

由上述得到储油罐发生变位时体积关于 h 的公式可以给出罐体变位后油位高度间隔为 10cm 的罐容表标定值（即进/出油量与罐内油位高度的表格）。

§ 4 模型的建立与求解

§ 4.1 问题一

§ 4.1.1 模型一

当储油罐体无变位时，储油罐圆柱体的接地一端为原点，以圆柱体高方向为 z 轴，建立笛卡尔坐标系，如图 4-1 所示，在高度为 h 时，利用微元法过垂直 z 轴的方向做截面 $S(z)$ ，对 $S(z)$ 关于 z 进行积分，得到体积关于 h 的公式。

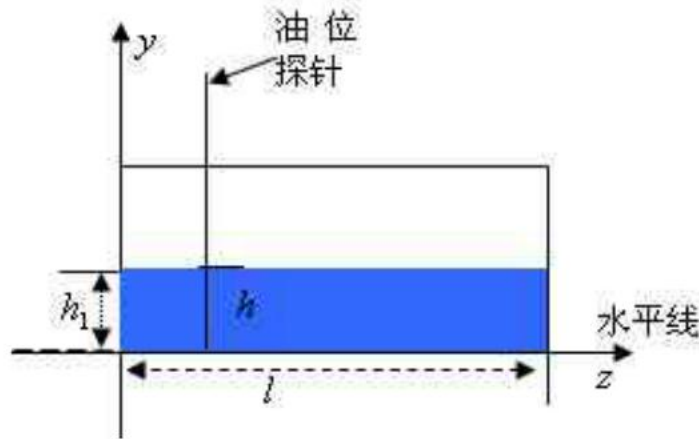


图 4-1

$$\text{由 } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow x = \frac{a}{b} \sqrt{b^2 - y^2}.$$

$$\begin{aligned} S(z) &= 2 \frac{a}{b} \int_0^h \sqrt{b^2 - y^2} dy \\ &= \frac{a}{b} \left[(h-b) \sqrt{2bh - h^2} + b^2 \arcsin \frac{h-b}{b} + \frac{1}{2} b^2 \pi \right] \end{aligned}$$

$$dV = S(z) dz$$

$$\begin{aligned} V &= \int_0^L S(z) dz \\ &= \frac{a}{b} L \left[(h-b) \sqrt{2bh - h^2} + b^2 \arcsin \frac{h-b}{b} + \frac{1}{2} \pi b^2 \right] \end{aligned}$$

$$V = \frac{a}{b} L \left[(h-b) \sqrt{2bh-h^2} + b^2 \arcsin \frac{h-b}{b} + \frac{1}{2} \pi b^2 \right]$$

利用 Matlab 中的命令 subs 代入附件 1 实验数据中的各高度得到储油罐中的理论油量 V 。

由附件 1 实验数据中进油量、出油量及储油罐罐内油量初值可以得到储油罐中的实际储油量，根据理论油量及实际油量就可以得出误差。

由附录中的程序 youliang1，我们得到了理论储油量，误差及相对误差。

进油后理论储油量与实际储油量随高度的变化规律如图 4-2 所示：

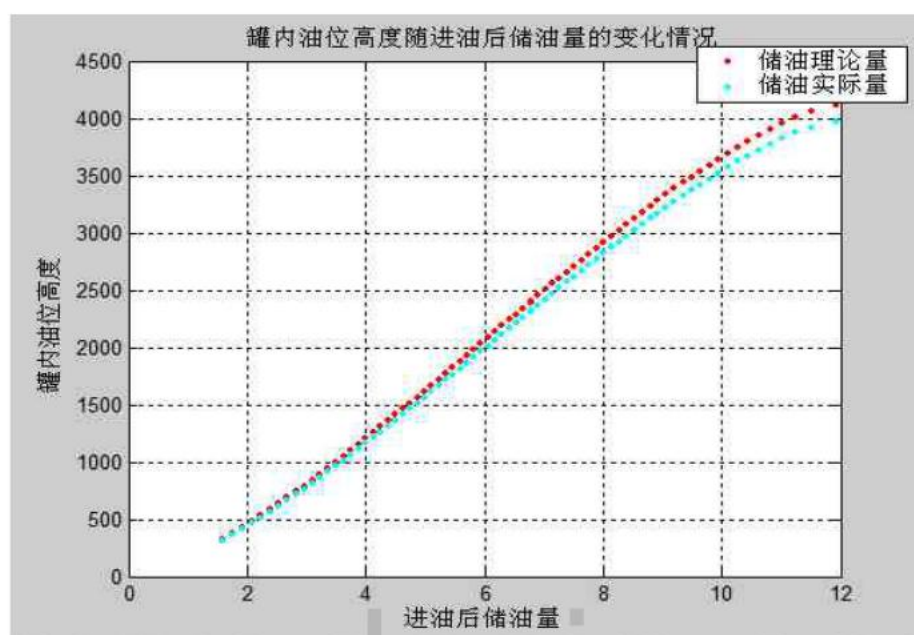


图 4-2

出油后理论储油量与实际储油量随高度的变化规律如图 4-3 所示：

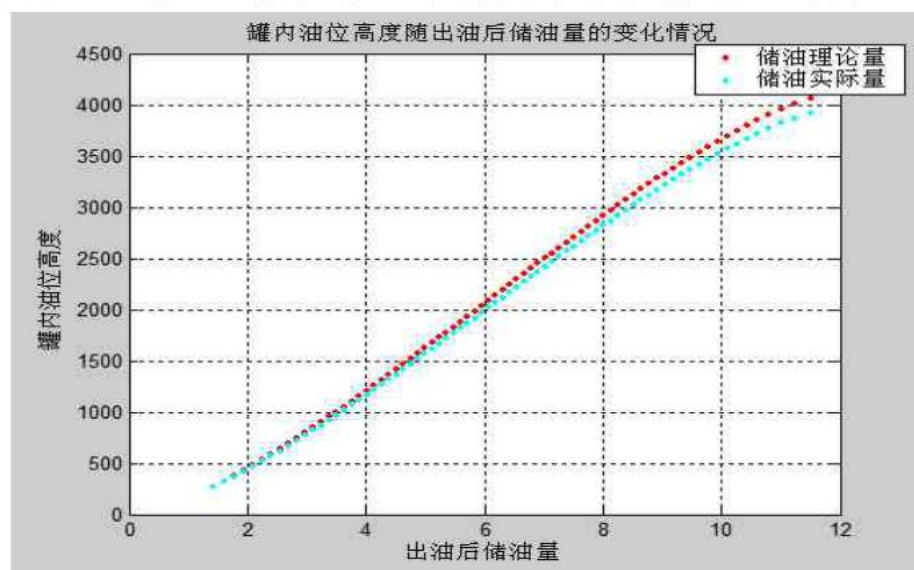


图 4- 3

无变位进油和无变位出油的储油理论量和储油实际量及误差和相对误差的数据如表 4-1 所示:

表 4- 1

无变位进油				无变位出油			
储油量实际值	储油量理论值	ε_i (误差)	相对误差	储油量实际值	储油量理论值	ε_i (误差)	相对误差
312	322.9	10.88	3.37%	3916.2	4052.8	136.61	3.37%
362	374.6	12.63	3.37%	3866.2	4001.0	134.86	3.37%
412	426.4	14.36	3.37%	3816.2	3949.3	133.13	3.37%
462	478.1	16.13	3.37%	3766.2	3897.6	131.38	3.37%
512	529.9	17.85	3.37%	3716.2	3845.8	129.65	3.37%
562	581.6	19.61	3.37%	3666.2	3794.1	127.88	3.37%
612	633.4	21.35	3.37%	3616.2	3742.3	126.16	3.37%
662	685.1	23.08	3.37%	3566.2	3690.6	124.39	3.37%
712	736.8	24.85	3.37%	3516.2	3638.8	122.65	3.37%
762	788.6	26.58	3.37%	3466.2	3587.1	120.92	3.37%
812	840.3	28.33	3.37%	3416.2	3535.4	119.17	3.37%
862	892.1	30.06	3.37%	3366.2	3483.6	117.43	3.37%
912	943.8	31.80	3.37%	3316.2	3431.9	115.70	3.37%
962	995.5	33.54	3.37%	3266.2	3380.1	113.94	3.37%
1012	1047.3	35.30	3.37%	3216.2	3328.4	112.18	3.37%
1062	1099.1	37.05	3.37%	3166.2	3276.6	110.45	3.37%
1112	1150.8	38.81	3.37%	3116.2	3224.9	108.70	3.37%
1162	1202.6	40.55	3.37%	3066.2	3173.2	106.96	3.37%
1212	1254.3	42.29	3.37%	3016.2	3121.4	105.22	3.37%
1262	1306.0	44.03	3.37%	2966.2	3069.7	103.46	3.37%
1312	1357.8	45.77	3.37%	2916.2	3017.9	101.73	3.37%
1362	1409.5	47.49	3.37%	2866.2	2966.2	100.00	3.37%
1412	1461.2	49.24	3.37%	2816.2	2914.4	98.23	3.37%
1462	1513.0	50.98	3.37%	2766.2	2862.7	96.50	3.37%
1512	1564.7	52.74	3.37%	2716.2	2811.0	94.76	3.37%
1562	1616.5	54.49	3.37%	2666.2	2759.2	93.02	3.37%
1612	1668.2	56.24	3.37%	2616.2	2707.5	91.26	3.37%
1662	1720.0	57.98	3.37%	2566.2	2655.7	89.51	3.37%
1712	1771.7	59.73	3.37%	2516.2	2603.9	87.76	3.37%
1762	1823.5	61.46	3.37%	2466.2	2552.2	86.03	3.37%
1812	1875.2	63.19	3.37%	2416.2	2500.5	84.30	3.37%
1862	1927.0	64.95	3.37%	2366.2	2448.7	82.53	3.37%
1912	1978.7	66.68	3.37%	2316.2	2397.0	80.80	3.37%
1962	2030.4	68.43	3.37%	2266.2	2345.3	79.07	3.37%

2012	2082.2	70.20	3.37%	2216.2	2293.5	77.32	3.37%
2062	2134.0	71.95	3.37%	2166.2	2241.7	75.55	3.37%
2112	2185.7	73.67	3.37%	2116.2	2190.0	73.84	3.37%
2162	2237.4	75.43	3.37%	2066.2	2138.3	72.08	3.37%
2212	2289.2	77.16	3.37%	2016.2	2086.5	70.32	3.37%
2262	2340.9	78.89	3.37%	1966.2	2034.8	68.60	3.37%
2312	2392.7	80.67	3.37%	1916.2	1983.0	66.85	3.37%
2315.8	2396.6	80.78	3.37%	1866.2	1931.3	65.12	3.37%
2365.8	2448.4	82.54	3.37%	1816.2	1879.5	63.35	3.37%
2367.1	2449.6	82.56	3.37%	1766.2	1827.8	61.61	3.37%
2417.1	2501.4	84.34	3.37%	1716.2	1776.1	59.88	3.37%
2467.1	2553.1	86.05	3.37%	1666.2	1724.3	58.12	3.37%
2517.1	2604.9	87.82	3.37%	1616.2	1672.6	56.37	3.37%
2567.1	2656.6	89.53	3.37%	1566.2	1620.8	54.64	3.37%
2617.1	2708.3	91.28	3.37%	1516.2	1569.1	52.87	3.37%
2667	2760.0	93.03	3.37%	1466.2	1517.3	51.14	3.37%
2668.8	2761.9	93.11	3.37%	1416.2	1465.6	49.42	3.37%
2718.8	2813.7	94.83	3.37%	1366.2	1413.8	47.65	3.37%
2768.8	2865.4	96.59	3.37%	1316.2	1362.1	45.91	3.37%
2818.8	2917.2	98.34	3.37%	1266.2	1310.4	44.18	3.37%
2868.8	2968.9	100.09	3.37%	1216.2	1258.6	42.42	3.37%
2918.8	3020.7	101.84	3.37%	1166.2	1206.9	40.68	3.37%
2968.8	3072.4	103.58	3.37%	1116.2	1155.1	38.94	3.37%
3018.8	3124.1	105.31	3.37%	1066.2	1103.4	37.19	3.37%
3068.8	3175.9	107.06	3.37%	1016.2	1051.6	35.43	3.37%
3118.8	3227.6	108.80	3.37%	966.2	999.9	33.71	3.37%
3168.8	3279.4	110.55	3.37%	916.2	948.2	31.96	3.37%
3168.9	3279.5	110.55	3.37%	866.2	896.4	30.20	3.37%
3218.9	3331.2	112.27	3.37%	816.2	844.6	28.45	3.37%
3268.9	3382.9	114.03	3.37%	766.2	792.9	26.71	3.37%
3318.9	3434.7	115.76	3.37%	716.2	741.2	24.99	3.37%
3368.9	3486.4	117.52	3.37%	666.2	689.4	23.25	3.37%
3418.9	3538.2	119.26	3.37%	616.2	637.7	21.51	3.37%
3468.9	3589.9	121.01	3.37%	566.2	585.9	19.74	3.37%
3518.9	3641.7	122.76	3.37%	516.2	534.2	18.02	3.37%
3568.9	3693.4	124.51	3.37%	466.2	482.5	16.28	3.37%
3618.9	3745.1	126.23	3.37%	416.2	430.7	14.50	3.37%
3668.9	3796.9	127.98	3.37%	366.2	379.0	12.77	3.37%
3718.9	3848.6	129.74	3.37%	316.2	327.2	11.02	3.37%
3768.9	3900.4	131.48	3.37%	266.2	275.5	9.28	3.37%
3818.9	3952.1	133.23	3.37%				
3868.9	4003.9	134.95	3.37%				
3918.9	4055.6	136.70	3.37%				
3968.9	4107.4	138.45	3.37%				

由上述的表格可以得出相对误差稳定，不会随高度发生变化。

当储油罐体发生变位时，我们就油面及油浮子的位置与倾斜角度的变化情况分成将该问题三类，然后就每一类进行分析，找出了储油罐中油的体积与油浮子位置的函数关系，其立体图如图 4-4 所示：

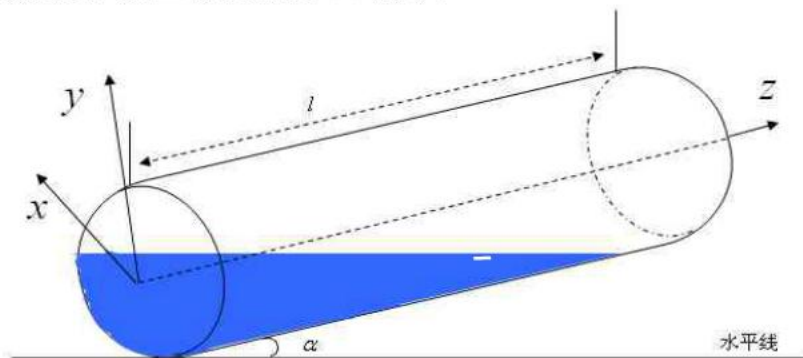


图 4-4

(1) 当油浮子的高度在 $0 \leq h \leq h_1 - z \tan \alpha$ 时，如图 4-5 所示，截面图如图 4-6 所示

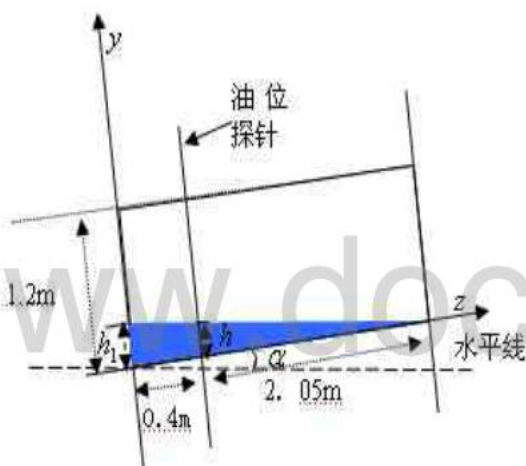


图 4-5

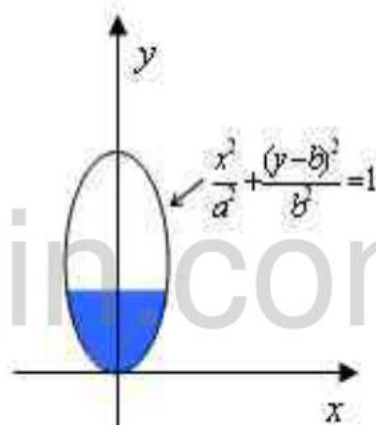


图 4-6 截面图

截面面积为：

$$\begin{aligned}
 S(z) &= \int_0^{h_1 - z \tan \alpha} 2 \frac{a}{b} \sqrt{b^2 - (y - b)^2} dy \\
 &= \frac{a}{b} \left((h_1 - z \tan(\alpha) - b) \sqrt{2b(h_1 - z \tan(\alpha)) - (h_1 - z \tan(\alpha))^2} \right. \\
 &\quad \left. + \frac{a}{b} \left(b^2 \arcsin \left(\frac{h_1 - z \tan(\alpha) - b}{b} \right) + \frac{1}{2} \pi b^2 \right) \right)
 \end{aligned}$$

所以油罐体积为：

$$\begin{aligned}
 V &= f(h_1) = \int_0^{\frac{h_1}{\tan \alpha}} S(z) dz \\
 &= \int_0^{\frac{h_1}{\tan \alpha}} \frac{a}{b} \left((h_1 - z \tan(\alpha) - b) \sqrt{2b(h_1 - z \tan(\alpha)) - (h_1 - z \tan(\alpha))^2} \right) \\
 &\quad + \int_0^{\frac{h_1}{\tan \alpha}} \frac{a}{b} \left(b^2 \arcsin \left(\frac{h_1 - z \tan(\alpha) - b}{b} \right) + \frac{1}{2} \pi b^2 \right) dz \quad (1.1)
 \end{aligned}$$

将 $h = h_1 - a \tan \alpha$ 代入公式 (1.1)

$$\begin{aligned}
 \text{令 } \frac{h + a_1 \tan \alpha - z \tan \alpha - b}{b} &= t, \text{ 则公式 (1.1)} \\
 V &= ab \frac{-b}{\tan \alpha} \int_{\frac{h + a_1 \tan \alpha - b}{b}}^{-1} (t \sqrt{1 - t^2} + \arcsin t + \frac{1}{2} \pi) dt \\
 &= \frac{ab^2}{\tan \alpha} \left(\frac{h + a_1 \tan \alpha - b}{b} \arcsin \left(\frac{h + a_1 \tan \alpha - b}{b} \right) + \sqrt{1 - \left(\frac{h + a_1 \tan \alpha - b}{b} \right)^2} \right) \\
 &\quad + \frac{ab^2}{\tan(\alpha)} \left(\frac{1}{2} \pi \left(\frac{h + a_1 \tan(\alpha) - b}{b} \right) - \frac{1}{3} \left(1 - \left(\frac{h + a_1 \tan(\alpha) - b}{b} \right)^2 \right)^{3/2} \right)
 \end{aligned}$$

(2) 当油浮子的高度在 $l \tan \alpha < h < 2b - a_1 \tan \alpha$ 时，如图 4-7 所示，

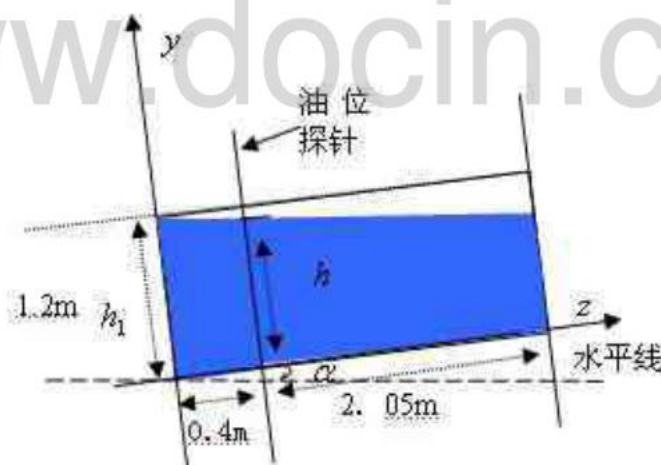


图 4-7

截面面积为：

$$\begin{aligned}
 S(z) &= \int_0^{h_1 - z \tan \alpha} 2 \frac{a}{b} \sqrt{b^2 - (y - b)^2} dy \\
 &= \frac{a}{b} \left((h_1 - z \tan(\alpha) - b) \sqrt{2b(h_1 - z \tan(\alpha)) - (h_1 - z \tan(\alpha))^2} \right) \quad \text{错误! 未找到引用源。} \\
 &\quad + \frac{a}{b} \left(b^2 \arcsin \left(\frac{h_1 - z \tan(\alpha) - b}{b} \right) + \frac{1}{2} \pi b^2 \right)
 \end{aligned}$$

到引用源。

所以油罐装油的体积为：

$$\begin{aligned}
 V &= \int_0^L S(z) dz \\
 &= ab \int_0^L \left(\frac{h + a_1 \tan(\alpha) - z \tan(\alpha) - b}{b^2} + \arcsin \left(\frac{h + a_1 \tan(\alpha) - z \tan(\alpha) - b}{b} \right) + \frac{1}{2} \pi \right) dz
 \end{aligned}$$

错误! 未找到引用源。

将 $h = h_1 - a_1 \tan(\alpha)$ 公式错误! 未找到引用源。 , 令 $\frac{h + a_1 \tan(\alpha) - z \tan(\alpha) - b}{b} = t$

公式 (2.2) 化为 :

$$V(h) = \frac{-ab^2}{\tan(\alpha)} \int_{(h + a_1 \tan(\alpha) - b)/b}^{(h + a_1 \tan(\alpha) - L \tan(\alpha) - b)/b} \left(t \sqrt{1 - t^2} + \arcsin(t) + \frac{1}{2} \pi \right) dt$$

将积分下限令为 p , 积分上限令为 q , 则

$$p = \frac{h + a_1 \tan(\alpha) - b}{b}, \quad q = \frac{h + a_1 \tan(\alpha) - L \tan(\alpha) - b}{b}$$

$$\begin{aligned}
 V(h) &= \frac{ab^2}{\tan(\alpha)} \left(\frac{1}{3} \left(\sqrt{1 - q^2} \right)^3 - \sqrt{1 - q^2} - q \sin(q) - \frac{1}{3} \left(\sqrt{1 - p^2} \right)^3 + p \sin(p) + \sqrt{1 - p^2} \right) \\
 &= \frac{ab^2}{\tan(\alpha)} \left[\frac{1}{3} \left(\sqrt{1 - \left(\frac{h + a_1 \tan(\alpha) - b}{b} \right)^2} \right)^3 \right]
 \end{aligned}$$

(3) 当油浮子的高度 $L \tan \alpha < h \leq 2b$ 时, 如图 4-8 所示,

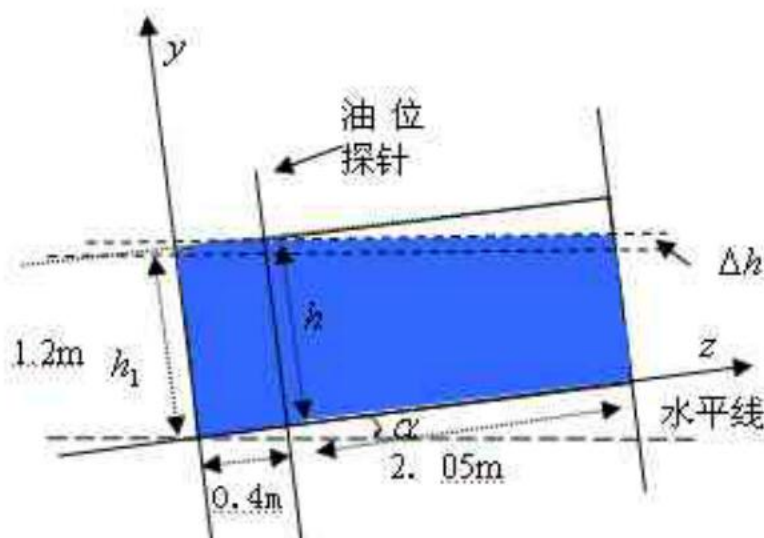


图 4-8

其中, $\begin{cases} \Delta h = h_1 - h = h_1 - 2b + a \tan \alpha, \\ h_2 = l \tan \alpha - \Delta h. \end{cases}$, 此时截面面积为:

$$\begin{aligned} S(z) &= \int_0^{h_2 - z \tan \alpha} 2 \frac{a}{b} \sqrt{b^2 - (y - b)^2} dy \\ &= \frac{a}{b} \left((h - (z - a_1) \tan(\alpha) - b) \sqrt{2b((h - (z - a_1) \tan(\alpha) - b)) - (h - (z - a_1) \tan(\alpha) - b)^2} \right) \\ &\quad + \frac{a}{b} \left(b^2 \arcsin \left(\frac{h - (z - a_1) \tan(\alpha) - b}{b} \right) + \frac{1}{2} \pi b^2 \right) \end{aligned}$$

所以油罐装油的体积为

$$V(h) = \int_L^{\frac{L \tan(\alpha) - h + 2b - a_1 \tan(\alpha)}{\tan(\alpha)}} s(z) dz$$

$$\text{令 } t = \frac{h - (z - a_1) \tan(\alpha) - b}{b}, p = \frac{h - (L - a_1) \tan(\alpha) - b}{b}, q = \frac{2h + 2a_1 \tan(\alpha) - 3b - L \tan(\alpha)}{b}$$

则上式公式变为:

$$\begin{aligned} V(h) &= \frac{-ab^2}{\tan(\alpha)} \int_p^q \left(t \sqrt{1-t^2} + \arcsin(t) + \frac{1}{2} \pi \right) dt \\ &= \frac{ab^2}{\tan(\alpha)} \left(\frac{1}{3} (\sqrt{1-q^2})^3 - \sqrt{1-q^2} - q \sin(q) - \frac{1}{3} (\sqrt{1-p^2})^3 + p \sin(p) + \sqrt{1-p^2} \right) \end{aligned}$$

利用 Matlab 中的命令 subs 代入附件 1 实验数据中的各高度得到储油罐中的理论油量 V 。

由附件 1 实验数据中进油量、出油量及储油罐罐内油量初值可以得到储油罐

中的实际油量，根据理论油量及实际油量就可以得出误差。

由附录中的程序 youliang2，我们得到了理论油量，误差及相对误差。

倾斜时进油后理论储油量与实际储油量随高度的变化规律如图 4-9 所示：

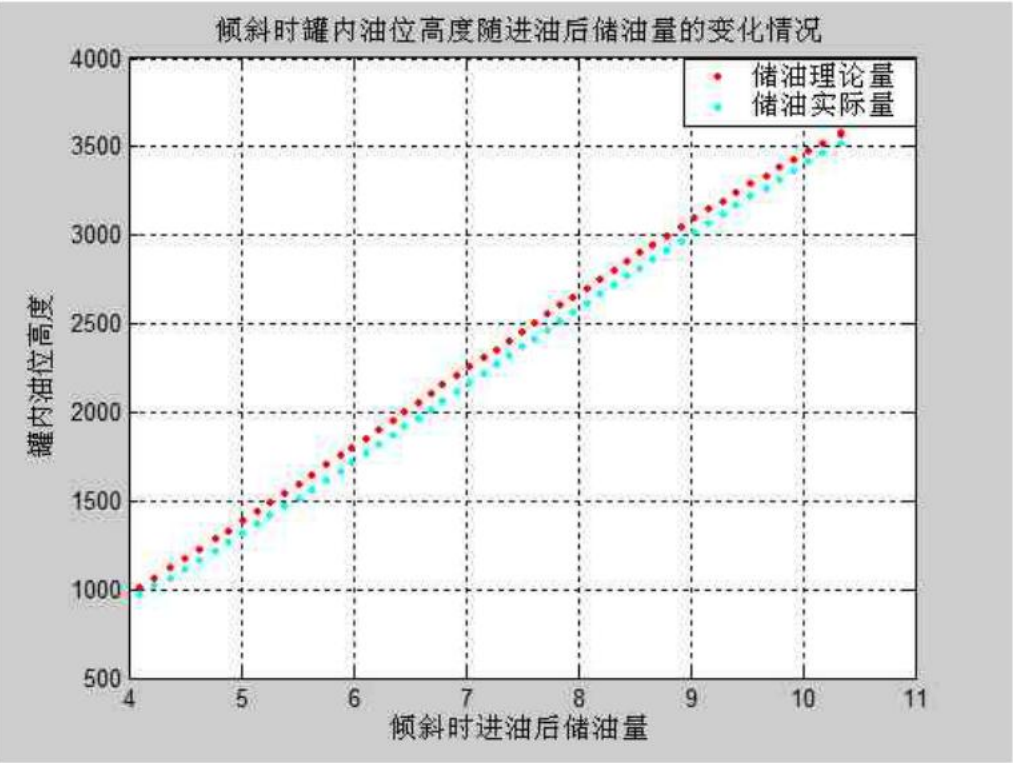


图 4- 9

倾斜时出油后理论储油量与实际储油量随高度的变化规律如图 4-10 所示：

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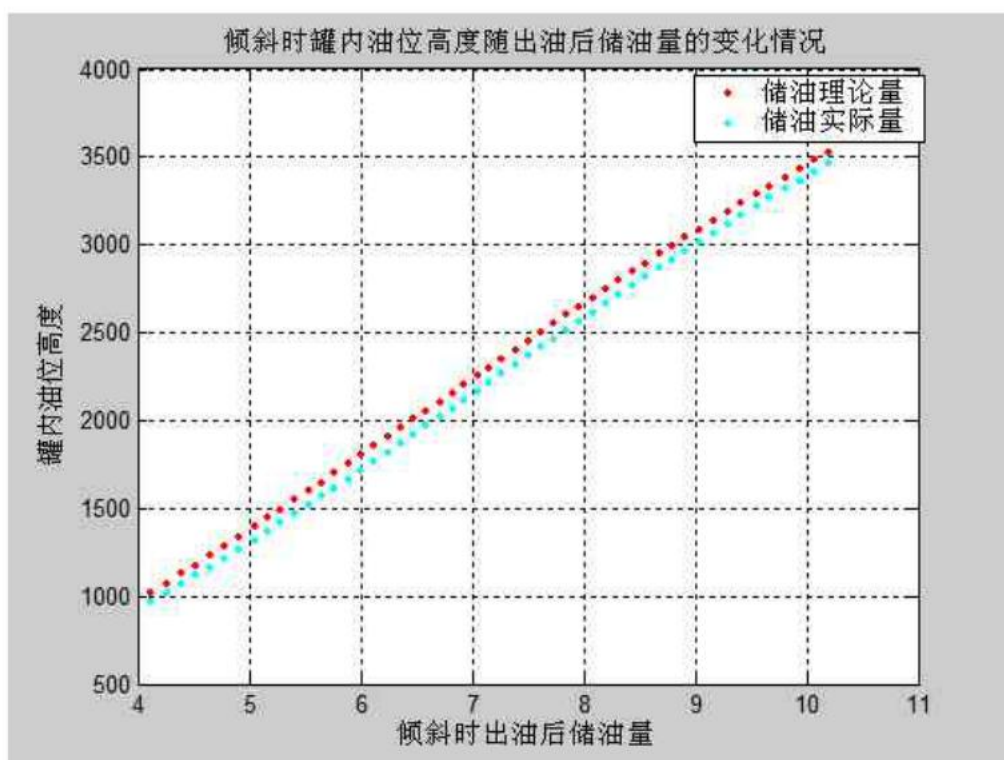


图 4-10

将倾斜进油和倾斜变位出油的储油理论值和储油实际值及误差和相对误差的数据如表 4-2 所示：

表 4-2

倾斜变位进油				倾斜变位出油			
储油量实际量	储油量理论值	ε_i (误差)	相对误差	储油量实际值	储油量理论值	ε_i (误差)	相对误差
962.9	1010.0	7.19	4.67%	3464.7	3523.1	58.37	1.66%
1012.9	1058.3	45.47	4.30%	3414.7	3478.1	63.35	1.82%
1062.9	1118.0	55.19	4.94%	3364.7	3430.4	65.67	1.91%
1112.9	1167.5	54.67	4.68%	3314.7	3382.0	67.28	1.99%
1162.9	1222.1	59.29	4.85%	3264.7	3330.9	66.19	1.99%
1212.9	1279.2	66.34	5.19%	3214.7	3289.4	74.69	2.27%
1262.9	1327.5	64.63	4.87%	3164.7	3234.5	69.77	2.16%
1312.8	1382.6	69.82	5.05%	3114.7	3188.9	74.17	2.33%
1362.8	1433.6	70.81	4.94%	3064.7	3137.3	72.57	2.31%
1412.7	1485.0	72.23	4.86%	3014.7	3088.9	74.11	2.40%
1462.7	1536.1	73.35	4.78%	2964.7	3040.1	75.40	2.48%
1512.7	1591.9	79.14	4.97%	2914.7	2989.2	74.48	2.49%
1562.7	1645.1	82.42	5.01%	2864.7	2947.7	82.98	2.82%
1612.7	1697.4	84.67	4.99%	2814.7	2891.1	76.36	2.64%
1662.7	1749.9	87.16	4.98%	2764.7	2845.4	80.62	2.83%

1712.7	1796.6	83.89	4.67%	2714.7	2794.0	79.31	2.84%
1762.7	1848.8	86.08	4.66%	2664.7	2747.5	82.72	3.01%
1812.7	1900.0	87.32	4.60%	2614.7	2695.8	81.07	3.01%
1862.7	1952.7	90.01	4.61%	2564.7	2644.5	79.78	3.02%
1912.7	1999.2	86.48	4.33%	2514.7	2598.1	83.33	3.21%
1962.7	2052.7	89.97	4.38%	2464.7	2547.1	82.38	3.23%
2012.7	2103.2	90.50	4.30%	2414.7	2500.2	85.47	3.42%
2062.7	2148.4	85.72	3.99%	2364.7	2451.9	87.11	3.55%
2112.7	2202.3	89.54	4.07%	2314.7	2402.9	88.14	3.67%
2162.7	2252.7	90.01	4.00%	2264.7	2349.7	84.97	3.62%
2212.7	2303.7	91.01	3.95%	2214.7	2298.8	84.11	3.66%
2262.7	2352.2	89.44	3.80%	2164.7	2256.0	91.29	4.05%
2312.7	2402.7	90.02	3.75%	2114.7	2204.4	89.66	4.07%
2362.7	2452.2	89.51	3.65%	2064.7	2156.6	91.83	4.26%
2412.7	2497.9	85.17	3.41%	2014.7	2106.7	91.97	4.37%
2462.7	2548.7	85.93	3.37%	1964.7	2053.1	88.35	4.30%
2512.7	2599.6	86.83	3.34%	1914.7	2005.6	90.81	4.53%
2562.7	2644.7	81.96	3.10%	1864.7	1953.5	88.78	4.54%
2612.7	2696.3	83.55	3.10%	1814.7	1905.1	90.39	4.74%
2662.7	2748.8	86.11	3.13%	1764.7	1852.8	88.02	4.75%
2712.7	2798.9	86.14	3.08%	1714.7	1801.5	86.77	4.82%
2762.7	2847.2	84.49	2.97%	1664.7	1752.7	87.99	5.02%
2812.7	2895.9	83.13	2.87%	1614.7	1699.3	84.55	4.98%
2862.7	2942.1	79.34	2.70%	1564.7	1645.9	81.18	4.93%
2912.7	2992.6	79.84	2.67%	1514.7	1602.0	87.27	5.45%
2962.7	3044.2	81.48	2.68%	1464.7	1544.1	79.35	5.14%
3012.7	3089.6	76.91	2.49%	1414.7	1492.6	77.89	5.22%
3062.7	3140.8	78.10	2.49%	1364.7	1444.1	79.41	5.50%
3112.7	3189.7	76.99	2.41%	1314.7	1392.3	77.56	5.57%
3162.7	3234.0	71.32	2.21%	1264.7	1333.4	68.62	5.15%
3212.7	3284.1	71.39	2.17%	1214.7	1280.5	65.78	5.14%
3262.7	3334.6	71.88	2.16%	1164.7	1230.6	65.91	5.36%
3312.7	3379.0	66.29	1.96%	1114.7	1175.1	60.37	5.14%
3362.7	3423.5	60.82	1.78%	1064.7	1124.7	59.97	5.33%
3412.7	3473.2	60.46	1.74%	1014.7	1067.8	53.10	4.97%
3462.7	3517.7	54.93	1.56%	964.7	1011.8	47.05	4.65%
3512.7	3569.5	56.73	1.59%				
3514.7	3573.2	58.49	1.64%				

由上述的程序还可以描绘出倾斜时储油量的误差随罐内油位高度的变化情况，如图 4-11 所示：

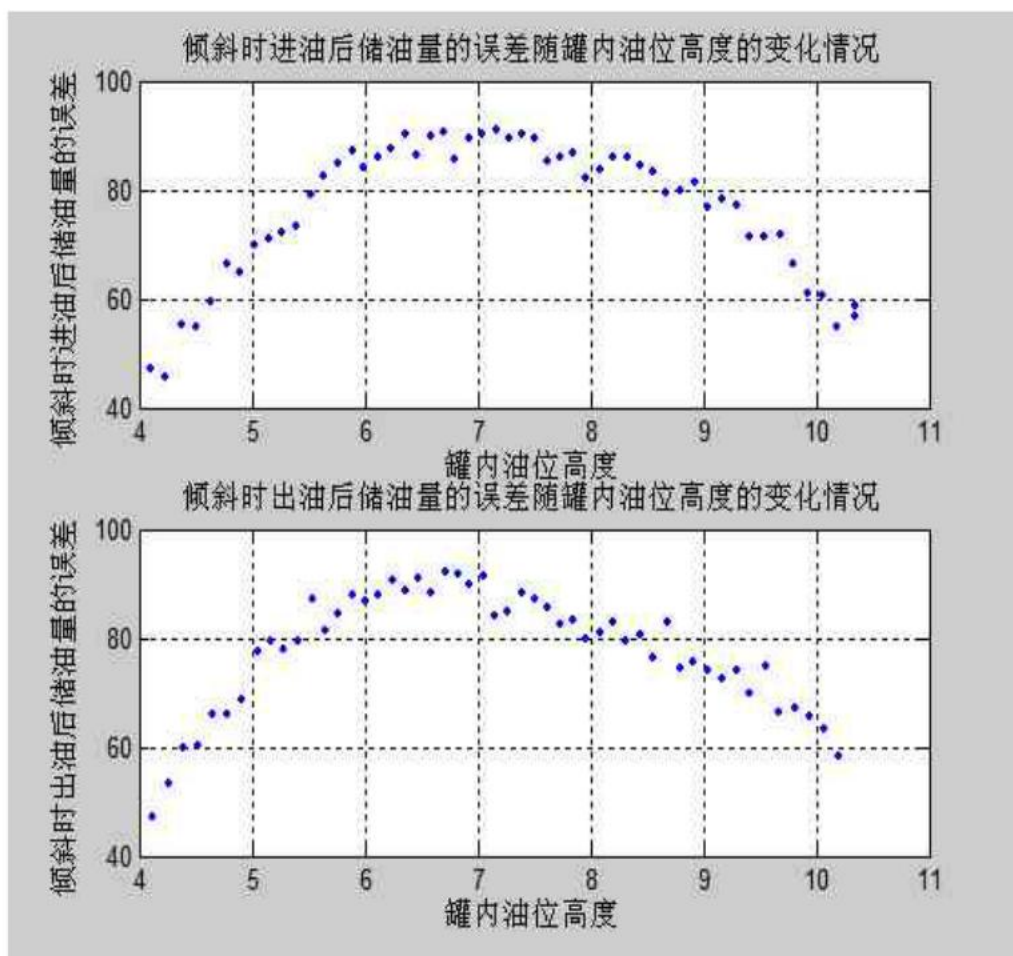


图 4- 11

由上述图我们可以知道倾斜时不管是进油后储油量的误差还是出油后储油量的误差均随罐内油位高度的变化呈正态分布，说明该模型建立得贴合实际。

由上述得到储油罐发生变位时体积关于 h 的公式我们可以给出罐体变位后油位高度间隔为 1cm 的罐容表标定值（即进/出油量与罐内油位高度的表格），如表 4-3 所示：

表 4- 3 罐体变位后油位高度间隔为 1cm 的罐容表标定值

高度 (h)	体积 (v)	高度 (h)	体积 (v)	高度 (h)	体积 (v)	高度 (h)	体积 (v)
cm	L	cm	L	cm	L	cm	L
1	3.531	26	461.5	51	1413.9	76	2491.3
2	6.2635	27	494	52	1456	77	2534
3	9.9748	28	527.1	53	1498.4	78	2576.6
4	14.7563	29	560.9	54	1540.9	79	2619.1
5	20.6908	30	595.2	55	1583.5	80	2661.4
6	27.8542	31	630.1	56	1626.3	81	2703.6
7	36.3163	32	665.6	57	1669.2	82	2745.5

8	46.1424	33	701.5	58	1712.2	83	2787.2
9	57.3935	34	738	59	1755.3	84	2828.7
10	70.127	35	774.9	60	1798.5	85	2870
11	84.3968	36	812.2	61	1841.8	86	2911.1
12	100.2541	37	850	62	1885.1	87	2951.8
13	117.7475	38	888.2	63	1928.5	88	2992.3
14	136.923	39	926.7	64	1971.9	89	3032.5
15	157.8246	40	965.7	65	2015.4	90	3072.4
16	180.4939	41	100.5	66	2058.8	91	3112
17	204.971	42	1044.6	67	2102.3	92	3151.2
18	228.9	43	1084.5	68	2145.7	93	3190.1
19	254.9	44	11248	69	2189.1	94	3228.6
20	281.9	45	1165.3	70	2232.5	95	3266.7
21	309.8	46	1206.2	71	2275.8	96	3304.4
22	338.5	47	1247.2	72	2319.1	97	3341.7
23	368.1	48	1.2886	73	2362.3	98	3378.5
24	398.5	49	1330.1	74	2405.4	99	3414.9
25	429.7	50	1371.9	75	2448.4	100	3450.7

§ 4.2 问题二

我们一开始考虑当储油罐水平放置时油量与浮油子高度的函数关系。

当储油罐水平放置无横向与纵向倾斜时，设油高为 h ，所建立的空间直角建立坐标系如图 4-12 所示：

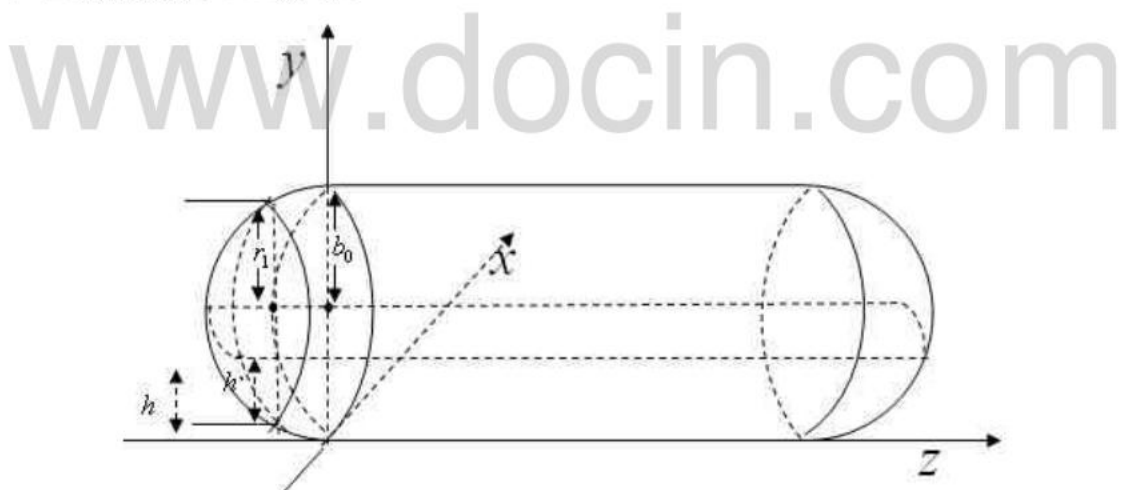


图 4-12

储油罐中储油的体积可分为两部分，圆柱体中储存油的体积与两端的球罐体中油的体积，根据对称性，两端的球罐体中油的体积相等。用垂直于 z 轴的平面截储油罐，根据截面法求体积公式可求出储油的体积。

第一种情况：当 $0 \leq h \leq b_0$ 时

1、对于圆柱体部分：截面圆的方程为 $x^2 + (y - b_0)^2 = b_0^2$

$$\text{由 } x^2 + (y - b_0)^2 = b_0^2 \Rightarrow x = \sqrt{b_0^2 - (y - b_0)^2}$$

$$\begin{aligned} \text{截面 } s(z) &= 2 \int_0^h \sqrt{b_0^2 - (y - b_0)^2} dy \\ &= 2 \left[\left(\frac{y - b_0}{2} \sqrt{b_0^2 - (y - b_0)^2} + \frac{b_0^2}{2} \arcsin \left(\frac{y - b_0}{b_0} \right) + \frac{1}{4} \pi b_0^2 \right) \right]_0^h \\ &= (h - b_0) \sqrt{b_0^2 - (h - b_0)^2} + b_0^2 \arcsin \left(\frac{h - b_0}{b_0} \right) + \frac{1}{2} \pi b_0^2 \end{aligned}$$

$$\begin{aligned} \text{体积 } v_1 &= \int_0^{L_0} s(z) dz \\ &= L_0 (h - b_0) \sqrt{b_0^2 - (h - b_0)^2} + L_0 b_0^2 \arcsin \left(\frac{h - b_0}{b_0} \right) + \frac{1}{2} \pi b_0^2 L_0 \end{aligned}$$

2、对于两端的球冠体，根据对称性，两端的球冠体储油体积相等，如图 4-13 所示：

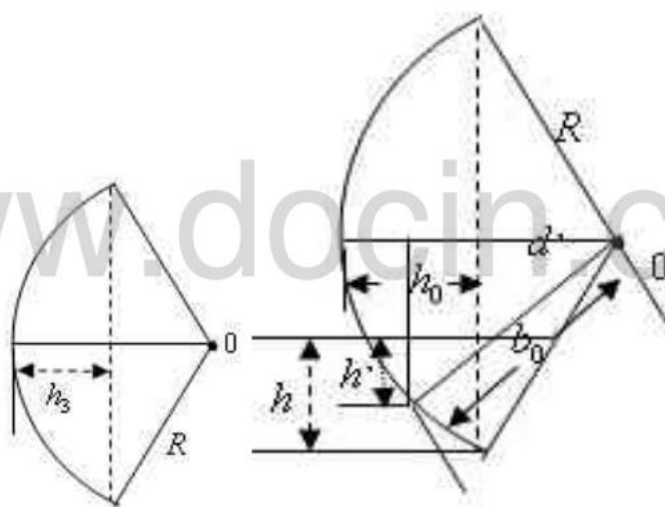


图 4-13

图 4-14

由图 4-13 可看出，根据几何关系：

$$R^2 = (R - h_3)^2 + b_0^2 \Rightarrow R = \frac{h_3^2 + b_0^2}{2h_3}$$

用平面 $z = z$ 截球罐体，设截得小圆半径为 r_1 ，所图 4-14 所示，根据几何关

系得:

$$r_1^2 = R^2 - (d' + z)^2 \quad \Rightarrow \quad d' = \sqrt{R^2 - b_0^2}$$

$$r_1^2 = R^2 - \left(\sqrt{R^2 - b_0^2} + z \right)^2 \quad r_1 = \sqrt{R^2 - \left(\sqrt{R^2 - b_0^2} + z \right)^2}$$

截面圆的油高为 h' , $h' = r_1 - (b_0 - h)$

$$\begin{aligned} \text{截面 } S(z) &= 2 \int_0^{h'} \sqrt{r_1^2 - (y - r_1)^2} dy \\ &= (h' - r_1) \sqrt{r_1^2 - (h' - r_1)^2} + r_1^2 \arcsin \left(\frac{h' - r_1}{r_1} \right) + \frac{1}{2} \pi r_1^2 \end{aligned}$$

$$h_3 = R - \sqrt{R^2 - b_0^2}$$

$$\begin{aligned} V_2 &= 2 \int_0^{h_3 - h_3 \frac{h}{b_0}} S(z) dz \\ &= 2 \int_0^{h_3 - h_3 \frac{h}{b_0}} (h - b_0) \sqrt{R^2 - \left(\sqrt{R^2 - b_0^2} + z \right)^2} - (h - b_0)^2 dz \\ &\quad + 2 \left[\int_0^{h_3 - h_3 \frac{h}{b_0}} \left(R^2 - \left(\sqrt{R^2 - b_0^2} + z \right)^2 \right) \arcsin \left(\frac{h - b_0}{\sqrt{R^2 - \left(\sqrt{R^2 - b_0^2} + z \right)^2}} \right) dz \right. \\ &\quad \left. + \pi \int_0^{h_3 - h_3 \frac{h}{b_0}} R^2 - \left(\sqrt{R^2 - b_0^2} + z \right)^2 dz \right] \end{aligned}$$

积分解得:

$$\begin{aligned} V_2 &= \frac{\pi h R^2}{2} - \frac{\pi h^3}{6} - \frac{2h}{3} \sqrt{2h_3 R - h^2 - h_3^2} (h_3 - R) - \frac{h_3^3 - 3h_3^2 R + 2R^3}{3} \arctan \left(\frac{h}{\sqrt{2h_3 R - h^2 - h_3^2}} \right) \\ &\quad + \frac{2}{3} R^3 \arctan \left(\frac{h(R - h_3)}{\sqrt{2h_3 R - h^2 - h_3^2}} \right) \end{aligned}$$

$$\begin{aligned}
V &= V_1 + 2V_2 \\
&= \pi h R^2 - \frac{\pi h^3}{3} + h L_0 \sqrt{b_0^2 - h^2} - \frac{4}{3} h \sqrt{b_0^2 - h^2} \sqrt{R^2 - b_0^2} \\
&\quad + \left[L_0 b_0^2 - \frac{2}{3} \sqrt{R^2 - b_0^2} (2R^2 + b_0^2) \right] \arctan \left(\frac{\sqrt{R^2 - b_0^2}}{\sqrt{b_0^2 - h^2}} \right) + \frac{4}{3} R^3 \arctan \left(\frac{h \sqrt{R^2 - b_0^2}}{R \sqrt{b_0^2 - h^2}} \right)
\end{aligned}$$

第二种情况：当 $b_0 \leq h \leq 2b_0$ 时

对于圆柱体的部分同第一种情况

$$\begin{aligned}
V_1 &= \int_0^{L_0} S(z) dz \\
&= L_0 (h - b_0) \sqrt{b_0^2 - (h - b_0)^2} + L_0 b_0^2 \arcsin \left(\frac{h - b_0}{b_0} \right) + \frac{1}{2} \pi b_0^2 L_0
\end{aligned}$$

对于球罐体体积

截面圆的油高为 h' $h' = r_1 + (h - b_0)$

$$\begin{aligned}
S(z) &= 2 \int_0^{h'} \sqrt{r_1^2 - (y - r_1)^2} dy \\
&= (h' - r_1) \sqrt{r_1^2 - (h' - r_1)^2} + r_1^2 \arcsin \left(\frac{h' - r_1}{r_1} \right) + \frac{1}{2} \pi r_1^2
\end{aligned}$$

这种情况与第一种情况相同，因此与第一种情况相同。

模型的建立：

V 与 h 的函数关系为：

$$\begin{aligned}
V &= V_1 + 2V_2 \\
&= \left[L_0 b_0^2 - \frac{2}{3} \sqrt{R^2 - b_0^2} (2R^2 + b_0^2) \right] \arctan \left(\frac{\sqrt{R^2 - b_0^2}}{\sqrt{b_0^2 - h^2}} \right) + \frac{4}{3} R^3 \arctan \left(\frac{h \sqrt{R^2 - b_0^2}}{R \sqrt{b_0^2 - h^2}} \right) \\
&\quad + \pi h R^2 - \frac{\pi h^3}{3} + h L_0 \sqrt{b_0^2 - h^2} - \frac{4}{3} h \sqrt{b_0^2 - h^2} \sqrt{R^2 - b_0^2}
\end{aligned}$$

我们就油所占的体积进行初步讨论,建立笛卡尔坐标系如图 4-15,可以得出球冠体的球面方程为:

$$x^2 + y^2 + \left[z - \left(h_3 - \frac{b_0^2 - h_0^2}{2h_0} \right) \right]^2 = \left[\frac{b_0^2 + h_0^2}{2h_0} \right]^2$$

圆柱体部分的方程： $x^2 + y^2 = b_0^2$ ；

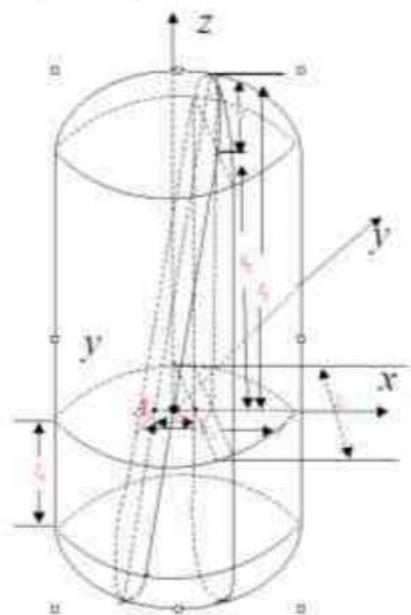


图 4-15

如图 4-15 所示，油罐体中所含油的部分分成两份，分别计算它们的体积。

$$\text{因为 } \tan \alpha = \frac{x_0 + h_1' - b_0}{l_2} = \frac{x_0 + h_1' - b_0}{h_3 + \frac{b_0 - x_0}{b_0} h_0}$$

$$\Rightarrow \tan \alpha = \frac{x_0 + h_1' - 1.5}{7 + \frac{x_0}{1.5}}$$

$$\Rightarrow x_0 + h_1' - 1.5 = \left(7 + \frac{x_0}{1.5} \right) \tan \alpha$$

$$\Rightarrow 1.5 x_0 + 1.5 h_1' - 2.25 = (10.5 + x_0) \tan \alpha = 10.5 \tan \alpha + x_0 \tan \alpha$$

$$\Rightarrow 1.5 h_1' - 2.25 - 10.5 \tan \alpha = (\tan \alpha - 1.5) x_0$$

$$\Rightarrow x_0 = \frac{1.5 h_1' - 2.25 - 10.5 \tan \alpha}{\tan \alpha - 1.5}$$

$$h_1 = (h_1' - b_0) \cos \beta + b_0 \text{ 且 } \cos \beta_{\max} = \frac{b_0 - h_1}{b_0} \Rightarrow \beta \leq \arccos \frac{b_0 - h_1}{b_0}$$

右半部分的体积 V_1 利用投影法可以由以下约束公式算出

$$\begin{cases} x^2 + y^2 + \left[z - \left(h_3 - \frac{b_0^2 - h_0^2}{2h_0} \right) \right]^2 = \left[\frac{b_0^2 + h_0^2}{2h_0} \right]^2 \\ x^2 + y^2 \leq b_0^2 \\ z \geq h_3 - \frac{b_0^2 - h_0^2}{2h_0} \\ x \geq x_0 \end{cases}$$

由上述第一个公式可以得出曲顶柱体的盖面方程为

$$\begin{aligned} z &= \sqrt{x^2 + y^2 - \left[\frac{b_0^2 + h_0^2}{2h_0} \right]^2} + h_3 - \frac{b_0^2 - h_0^2}{2h_0} \\ \Rightarrow V_1 &= \iiint_{\substack{x^2 + y^2 \leq b_0^2 \\ x \geq x_0}} \sqrt{x^2 + y^2 - \left[\frac{b_0^2 + h_0^2}{2h_0} \right]^2} + h_3 - \frac{b_0^2 - h_0^2}{2h_0} dx dy \\ &= \iiint_{\substack{x^2 + y^2 \leq 2.25 \\ x \geq x_0}} \sqrt{x^2 + y^2 - \left[\frac{3.35}{3} \right]^2} + 6 - \frac{1.25}{2} dx dy \\ &= \iint_{\substack{x^2 + y^2 \leq 2.25 \\ x \geq x_0}} \sqrt{x^2 + y^2 - 1.25} + 6.625 dx dy \end{aligned}$$

左半部分的体积 V_2 近似利用截面法可以由以下公式算出，由图 4-15 可以得出 A 点的坐标为 $(-(h_1' - b_0), 0, 0)$ ；过垂直于 x 轴做截面，设过 x 的截面面积为

$$S(x)；由 x^2 + y^2 = b_0^2 \Rightarrow y = \pm \sqrt{b_0^2 - x^2} \Rightarrow l_4 = 2|y| = 2\sqrt{b_0^2 - x^2}$$

$$\text{由图 4-15 可知 } \frac{x + h_1' - b_0}{x_0 + h_1' - b_0} = \frac{h_2}{l_2} = \frac{h_2}{h_3 + l_3} \text{ 而且 } \frac{l_3}{h_0} = \frac{b_0 - x_0}{b_0} \Rightarrow l_3 = \frac{b_0 - x_0}{b_0} h_0$$

$$\Rightarrow h_2 = \frac{x + h_1' - b_0}{x_0 + h_1' - b_0} \left(h_3 + \frac{b_0 - x_0}{b_0} h_0 \right)$$

过 x 的截面面积为 $S(x)$ 近似可表示如下:

$$\Rightarrow S(x) = \frac{1}{2} l_4 h_2 = \sqrt{b_0^2 - x^2} \frac{x + h_1' - b_0}{x_0 + h_1' - b_0} \left(h_3 + \frac{b_0 - x_0}{b_0} h_0 \right)$$

$$= \sqrt{2.25 - x^2} \frac{x + h_1' - 1.5}{x_0 + h_1' - 1.5} \left(6 + \frac{1.5 - x_0}{1.5} \right)$$

$$= \sqrt{2.25 - x^2} \frac{x + h_1' - 1.5}{x_0 + h_1' - 1.5} \left(7 - \frac{x_0}{1.5} \right)$$

$$\begin{aligned} V_2 &= \int_{1.5-h_1'}^{x_0} S(x) dx \\ &= \int_{1.5-h_1'}^{x_0} \sqrt{2.25 - x^2} \frac{x + h_1' - 1.5}{x_0 + h_1' - 1.5} \left(7 - \frac{x_0}{1.5} \right) dx \end{aligned}$$

有上述的 V_1, V_2 可以得到储油罐中油量与油位高度及变位参数的函数关系:

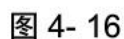
$$\begin{aligned} V &= V_1 + V_2 \\ &= \iiint_{\substack{x^2 + y^2 \leq 2.25 \\ x \geq x_0}} \sqrt{x^2 + y^2 - 1.25} + 6.625 dx dy + \int_{1.5-h_1'}^{x_0} \sqrt{2.25 - x^2} \frac{x + h_1' - 1.5}{x_0 + h_1' - 1.5} \left(7 - \frac{x_0}{1.5} \right) dx \end{aligned}$$

以下为在固定的横向偏转角度 β 条件下,就纵向倾斜角度 α 的变化进行分类讨论。

令 $2a = 3m, a_2 = 2m, a_3 = 1m$;

情形 I . 当 $\alpha \leq \arctan \frac{2a}{l_1}$ 时,

(1) 当 $h_1 \leq l_1 \tan \alpha$ 时,该情况的图 4-16 所示:



$$V_1 = \int_0^{\frac{k_1}{\tan \alpha}} S(z) dz = \int_0^{\frac{k_1}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

令 $A = 1.5\text{ m}$, $B = 1.5\text{ m}$, $C = 1\text{ m}$.

$$\frac{x^2}{A^2} + \frac{(y-B)^2}{B^2} = 1 \Rightarrow x = \frac{A}{B} \sqrt{2By - y^2}$$

$$\frac{(y - B)^2}{B^2} + \frac{z^2}{C^2} = 1 \Rightarrow z = \frac{C}{B} \sqrt{2By - y^2}$$

$$\begin{aligned} V_0 &= \int_0^{h_1} (2xl_1 + \pi xz) dy \\ &= 2l \int_0^{h_1} x dy + \pi \int_0^{h_1} xz dy \end{aligned}$$

x 和 z 式代入 V_0 式, 得:

$$V_0 = 2 \int_0^{h_1} \frac{A}{B} \sqrt{B^2 + (y - B)^2} dy + \pi \int_0^{h_1} \frac{A}{B} \sqrt{2By - y^2} \frac{C}{B} \sqrt{2By - y^2} dy$$

$$= AB \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[Bh_1^2 - \frac{1}{3} h_1^3 \right]$$

$$V_2 = \frac{V_0}{2} = \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

由此可得：

$$\begin{aligned} V &= V_1 + V_2 \\ &= \int_0^{\frac{h_1}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz + \\ &\quad \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right] \end{aligned}$$

(2) 当 $l_1 \tan \alpha \leq h_1 \leq 2a$ 时，该情况如图 4-17 所示：

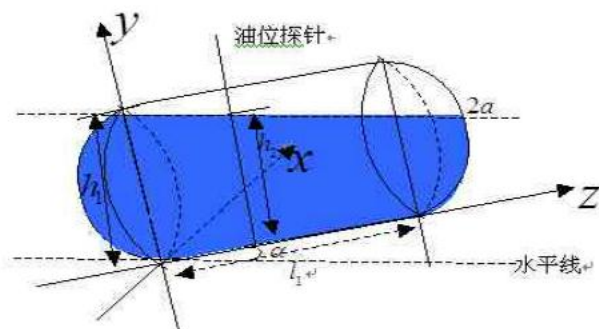


图 4-17

$$S(z) = \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$V_1 = \int_0^{l_1} S(z) dz = \int_0^{l_1} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

同上可得

$$V_2 = \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

$$\begin{aligned} V &= V_1 + V_2 \\ &= \int_0^{l_1} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz + \\ &\quad \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right] \end{aligned}$$

(3) 当 $h_1 = 2a$ 且 $h \leq 2a$ 时, 该情况如图 4-18 所示:

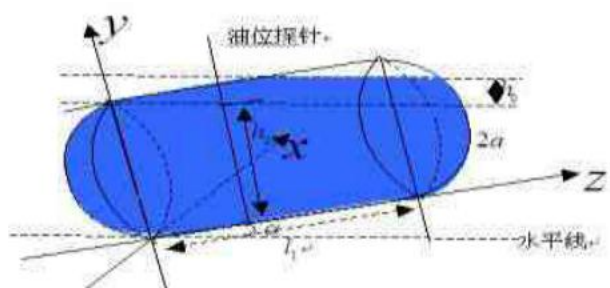


图 4-18

$$\text{令 } h_0 = l_1 \tan \alpha, \Delta h = h - h_2$$

$$\frac{l_1 - a_2}{l_1} = \frac{h_2}{2a} \Rightarrow h_2 = \frac{2a(l_1 - a_2)}{l_1}, h_1' = h_0 - \Delta h = l_1 \tan \alpha - h + \frac{2a(l_1 - a_2)}{l_1}$$

$$\frac{h_1' \tan \alpha - z}{h_1' \tan \alpha} = \frac{y_0'}{h_1'}, S(z) = \int_0^{y_0'} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$\Rightarrow V_1' = \int_0^{\frac{h_1'}{\tan \alpha}} S(z) dz = \int_0^{\frac{h_1'}{\tan \alpha}} \int_0^{y_0'} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$V_2' = \frac{AB}{2} \left[\frac{h_1' - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1' - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1'^2 - \frac{1}{3} h_1'^3 \right]$$

设实际储油罐总体积为 $V_{\text{全}}$,

$$\begin{aligned} V_{\text{全}} &= \pi R^2 L_0 + \frac{2\pi (R - h_0)^2 (3b_0 - R - h_0)}{3} \\ &= \pi R^2 L_0 + \frac{2\pi (b_0 - \sqrt{b_0^2 - R^2})^2 (3b_0 - b_0 + \sqrt{b_0^2 - R^2})}{3} \\ &= \pi R^2 L_0 + \frac{2\pi}{3} (b_0 - \sqrt{b_0^2 - R^2})^2 (2b_0 + \sqrt{b_0^2 - R^2}) \end{aligned}$$

$$\begin{aligned}
\Rightarrow V &= V_{\text{全}} - V_1' - V_2' \\
&= V_{\text{全}} - \int_0^{\frac{h_1'}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz \\
&\quad - \frac{AB}{2} \left[\frac{h_1' - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1' - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1'^2 - \frac{1}{3} h_1'^3 \right]
\end{aligned}$$

情形 II. 当 $\arctan \frac{2a}{l_1} < \alpha \leq \arctan \frac{l_1 - 2a}{2a}$ 时,

(1) 当 $h_1 \leq 2a$ 时, 该情况的图示如下:

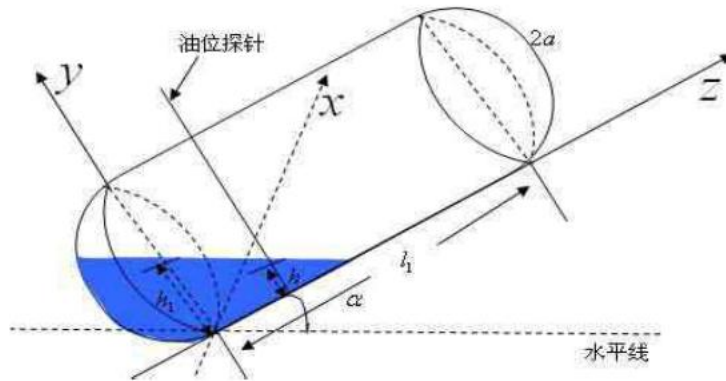


图 4-19

$$\frac{h}{h_1} = \frac{l_1 - a_2}{l_2}, \quad \frac{h_1 \tan \alpha - z}{h_1 \tan \alpha} = \frac{y_0}{h_1}, \quad S(z) = \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$V_1 = \int_0^{\frac{h_1}{\tan \alpha}} S(z) dz = \int_0^{\frac{h_1}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$V_2 = \frac{V_0}{2} = \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

$$V = V_1 + V_2$$

$$= \int_0^{\frac{h_1}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz +$$

$$\frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

(2) 当 $h_1 = 2a$ 且 $h \leq 2a - \frac{l_1 - a_2}{\tan \alpha}$ 时, 该情况如图 4-20 所示:

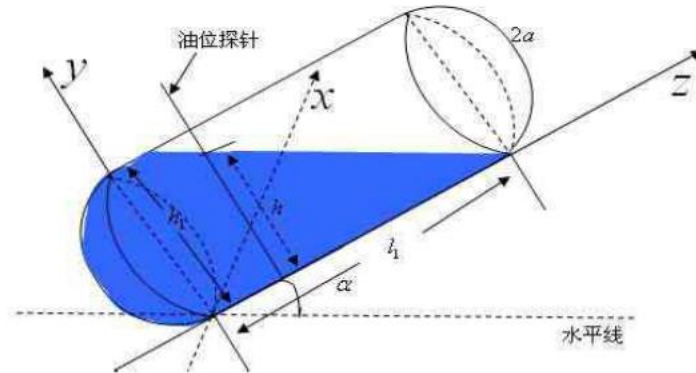


图 4-20

$$S(z) = \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$V_1 = \int_0^{l_1} S(z) dz = \int_0^{l_1} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$V_2 = \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

$$V = V_1 + V_2$$

$$= \int_0^{l_1} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz + \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

(3) 当 $2a - \frac{l_1 - a_2}{\tan \alpha} < h \leq 2a$ 时, 该情况如图 4-21 所示:

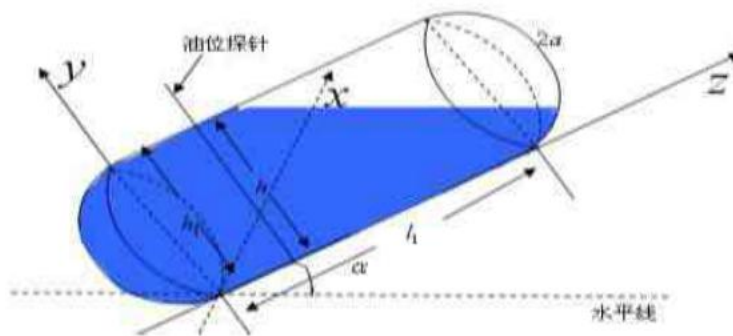


图 4-21

$$\text{令 } h_0 = l_1 \tan \alpha, \Delta h = h - h_2$$

$$\frac{l_1 - a_2}{l_1} = \frac{h_2}{2a} \Rightarrow h_2 = \frac{2a(l_1 - a_2)}{l_1}, h_1' = h_0 - \Delta h = l_1 \tan \alpha - h + \frac{2a(l_1 - a_2)}{l_1}$$

$$\frac{h_1' \tan \alpha - z}{h_1' \tan \alpha} = \frac{y_0'}{h_1'}, S(z) = \int_0^{y_0'} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$\Rightarrow V_1' = \int_0^{\frac{h_1'}{\tan \alpha}} S(z) dz = \int_0^{\frac{h_1'}{\tan \alpha}} \int_0^{y_0'} \int_0^{y_0'} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$V_2' = \frac{AB}{2} \left[\frac{h_1' - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1' - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1'^2 - \frac{1}{3} h_1'^3 \right]$$

$$\Rightarrow V = V_{\text{全}} - V_1' - V_2'$$

$$= V_{\text{全}} - \int_0^{\frac{h_1'}{\tan \alpha}} \int_0^{y_0'} \int_0^{y_0'} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$- \frac{AB}{2} \left[\frac{h_1' - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1' - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1'^2 - \frac{1}{3} h_1'^3 \right]$$

情形Ⅲ. 当 $\alpha > \arctan \frac{l_1 - 2a}{2a}$ 时,

(1) 当 $h_1 \leq 2a$ 时, 该情况如图 4-22 所示:

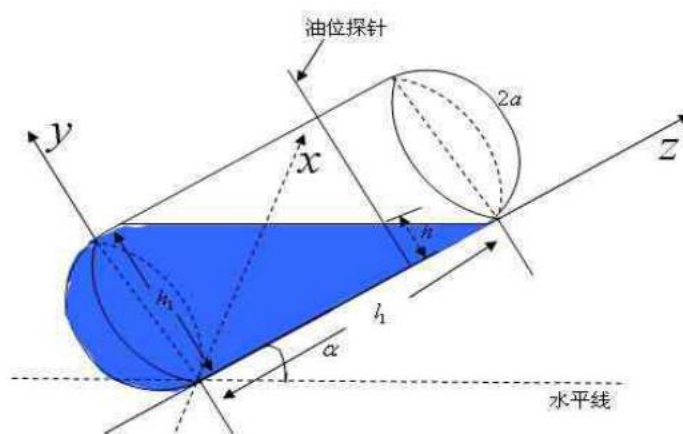


图 4-22

$$\frac{h}{h_1} = \frac{l_1 - a_2}{l_2}, \frac{h_1 \tan \alpha - z}{h_1 \tan \alpha} = \frac{y_0}{h_1}, S(z) = \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$V_1 = \int_0^{\frac{h_1}{\tan \alpha}} S(z) dz = \int_0^{\frac{h_1}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$V_2 = \frac{V_0}{2} = \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

$$V = V_1 + V_2$$

$$= \int_0^{\frac{h_1}{\tan \alpha}} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz +$$

$$\frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

(2) 当 $h_1 = 2a$ 且 $h \leq 2a$ 时, 该情况如图 4-23 所示:

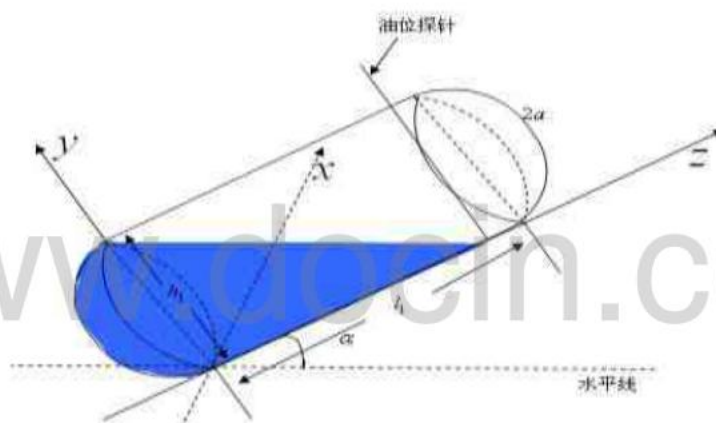


图 4-23

$$S(z) = \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy$$

$$V_1 = \int_0^{l_1} S(z) dz = \int_0^{l_1} \int_0^{y_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz$$

$$V_2 = \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]$$

$$\begin{aligned}
V &= V_1 + V_2 \\
&= \int_0^{l_1} \int_0^{x_0} 2(a \sin \beta + \sqrt{a^2 - (y - a \cos \beta)^2}) dy dz + \\
&\quad \frac{AB}{2} \left[\frac{h_1 - B}{B} \sqrt{1 - \left(\frac{h - B}{B} \right)^2} + \arcsin \frac{h_1 - B}{B} + \frac{\pi}{2} \right] + \frac{\pi AC}{B^2} \left[B h_1^2 - \frac{1}{3} h_1^3 \right]
\end{aligned}$$

§5 模型的进一步讨论

就第二问我们在上述讨论一般情况分析的基础上对左半部分进行了详细的分析。

$$S_1 : \begin{cases} x^2 + y^2 = b_0^2 \\ -(h_1 - b_0) \leq x \leq \operatorname{ctg} \alpha - (h_1 - b_0) \end{cases}, \quad z(x, y) : z = \operatorname{ctg} \alpha [x + (h_1 - b_0)]$$

$$\begin{aligned}
V_1 &= \iint_{\substack{x^2 + y^2 \leq b_0^2 \\ -(h_1 - b_0) \leq x \leq \operatorname{ctg} \alpha - (h_1 - b_0)}} \operatorname{ctg} \alpha [x + (h_1 - b_0)] dx dy \\
&= \operatorname{ctg} \alpha \iint_{\substack{x^2 + y^2 \leq b_0^2 \\ -(h_1 - b_0) \leq x \leq \operatorname{ctg} \alpha - (h_1 - b_0)}} x dx dy + \operatorname{ctg} \alpha (h_1 - b_0) S_1
\end{aligned}$$

$$\text{因 } \theta = \arcsin \frac{x_0}{b_0}$$

$$\text{所以阴影部分面积} = 2 \left(\frac{1}{2} b_0 x_0 \cos \theta + \theta b_0^2 \right) = b_0 x_0 \cos \theta + 2 \theta b_0^2$$

$$= b_0 x_0 \sqrt{1 - \frac{x_0^2}{b_0^2}} + 2 b_0^2 \arcsin \frac{x_0}{b_0} = x_0 \sqrt{b_0^2 - x_0^2} + 2 b_0^2 \arcsin \frac{x_0}{b_0}$$

$$\Rightarrow S_1 = (h_1 - b_0) \sqrt{b_0^2 - (h_1 - b_0)^2} + b_0^2 \arcsin \frac{h_1 - b_0}{b_0}$$

$$+ [\operatorname{ctg} \alpha - (h_1 - b_0)] \sqrt{b_0^2 - [\operatorname{ctg} \alpha - (h_1 - b_0)]^2} + 2 b_0^2 \arcsin \frac{\operatorname{ctg} \alpha - (h_1 - b_0)}{b_0}$$

$$\begin{aligned}
\text{因 } \iint_{\substack{x^2 + y^2 \leq b_0^2 \\ -(h_1 - b_0) \leq x \leq \operatorname{ctg} \alpha - (h_1 - b_0)}} x dx dy &= \int_{-(h_1 - b_0)}^{\operatorname{ctg} \alpha - (h_1 - b_0)} x dx \int_{-\sqrt{b_0^2 - x^2}}^{\sqrt{b_0^2 - x^2}} dy = \int_{-(h_1 - b_0)}^{\operatorname{ctg} \alpha - (h_1 - b_0)} \sqrt{b_0^2 - x^2} dx^2 \\
&= -\frac{2}{3} (b_0^2 - x^2)^{\frac{3}{2}} \Big|_{-(h_1 - b_0)}^{\operatorname{ctg} \alpha - (h_1 - b_0)} = \frac{2}{3} [b_0^2 - (h_1 - b_0)^2]^{\frac{3}{2}} - \frac{2}{3} [b_0^2 - [\operatorname{ctg} \alpha - (h_1 - b_0)]^2]^{\frac{3}{2}}
\end{aligned}$$

以下方程用来求 x_0 :

$$\sqrt{\left(\frac{b_0^2 + h_0^2}{2d}\right)^2 - x_0^2} + \left(l - \frac{b_0^2 - h_0^2}{2h_0}\right) = \operatorname{ctg} \alpha [x_0 + (h_1 - b_0)]$$

$$\Rightarrow \left(\frac{b_0^2 + h_0^2}{2h_0}\right)^2 - x_0^2 = \left(\operatorname{ctg} \alpha [x_0 + (h_1 - b_0)] - \left(l - \frac{b_0^2 - h_0^2}{2h_0}\right)\right)^2$$

$$\text{令 } \Delta = \operatorname{ctg} \alpha (h_1 - b_0) - \left(l - \frac{b_0^2 - h_0^2}{2h_0}\right) < 0$$

$$\Rightarrow \csc^2 \alpha x_0^2 + 2 \operatorname{ctg} \alpha \cdot \Delta \cdot x_0 - \left(\frac{b_0^2 + h_0^2}{2h_0}\right)^2 = 0$$

$$\Rightarrow x_0 = \frac{2 \operatorname{ctg} \alpha \cdot \Delta \pm \sqrt{4 \operatorname{ctg}^2 \alpha \cdot \Delta^2 - 4 \csc^2 \alpha \left(\frac{b_0^2 + h_0^2}{2h_0}\right)^2}}{2 \csc^2 \alpha}$$

$$= \frac{\operatorname{ctg} \alpha \cdot \Delta \pm \sqrt{\operatorname{ctg}^2 \alpha \cdot \Delta^2 - \csc^2 \alpha \left(\frac{b_0^2 + h_0^2}{2h_0}\right)^2}}{\csc^2 \alpha}$$

$$V_1 = \frac{2}{3} \operatorname{ctg} \alpha \left\{ \left[b_0^2 - (h_1 - b_0)^2 \right]^{\frac{3}{2}} - \left[b_0^2 - [\operatorname{ctg} \alpha - (h_1 - b_0)]^2 \right]^{\frac{3}{2}} \right\}$$

$$+ \operatorname{ctg} \alpha (h_1 - b_0) \left[(h_1 - b_0) \sqrt{b_0^2 - (h_1 - b_0)^2} + b_0^2 \arcsin \frac{h_1 - b_0}{b_0} + b_0^2 \arcsin \frac{\operatorname{ctg} \alpha - (h_1 - b_0)}{b_0} \right]$$

$$+ \operatorname{ctg} \alpha (h_1 - b_0) \left\{ [\operatorname{ctg} \alpha - (h_1 - b_0)] \sqrt{b_0^2 - [\operatorname{ctg} \alpha - (h_1 - b_0)]^2} \right\}$$

首先我们求出截面和球冠体的交线：

$$\begin{cases} \text{截面} & z = \operatorname{ctg} \alpha [x_0 + (h_1 - b_0)] \\ \text{球面方程} & x^2 + y^2 + \left[z - \left(h_3 - \frac{b_0^2 - h_0^2}{2h_0} \right) \right]^2 \leq \left[\frac{b_0^2 + h_0^2}{2h_0} \right]^2 \\ & z \geq l \end{cases}$$

§ 6 模型的优缺点

§ 6.1 优点

(1) 对于问题一, 从储油罐无变位与变位两个方面建立了模型, 求得的储油量理论值与储油量实际值吻合得较好, 相对误差小; 特别是水平时相对误差稳定, 符合要求, 因此能准确的预测出罐体变位后油位高度间隔为 1cm 的罐容表标定值。

(2) 对于问题二, 先就一般情况考虑, 再细分成 8 种情形, 思维缜密, 考虑周到。利用 Matlab 软件求出值, 求得各高度下的储油量理论值与储油量实际值吻合得较好, 相对误差小, 符合要求, 因此能准确的预测出罐体变位后油位高度间隔为 10cm 的罐容表标定值。

(3) 模型实用性好, 具有很强的现实应用指导意义, 数学推导严谨, 理论性强。

§ 6.2 缺点

模型的假设不够充分, 对误差的检验不够详细, 没有从多方面进行误差诊断。

§ 7 参考文献

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§ 7 附录

§ 7.1 附件 1

表 7-1 无变位进油

流水号	油罐号	累加进油量/L	油位高度/mm	采集时间
11	1	50	159.02	2010-08-18 10:32:18
12	1	100	176.14	2010-08-18 10:33:18

13	1	150	192.59	2010-08-18 10:34:18
14	1	200	208.5	2010-08-18 10:35:18
15	1	250	223.93	2010-08-18 10:36:18
16	1	300	238.97	2010-08-18 10:37:08
17	1	350	253.66	2010-08-18 10:38:08
18	1	400	268.04	2010-08-18 10:39:08
19	1	450	282.16	2010-08-18 10:40:08
20	1	500	296.03	2010-08-18 10:41:08
21	1	550	309.69	2010-08-18 10:41:58
22	1	600	323.15	2010-08-18 10:42:58
23	1	650	336.44	2010-08-18 10:43:58
24	1	700	349.57	2010-08-18 10:45:18
25	1	750	362.56	2010-08-18 10:46:08
26	1	800	375.42	2010-08-18 10:47:08
27	1	850	388.16	2010-08-18 10:48:08
28	1	900	400.79	2010-08-18 10:49:08
29	1	950	413.32	2010-08-18 10:49:58
30	1	1000	425.76	2010-08-18 10:50:58
31	1	1050	438.12	2010-08-18 10:51:58
32	1	1100	450.4	2010-08-18 10:52:48
33	1	1150	462.62	2010-08-18 10:53:48
34	1	1200	474.78	2010-08-18 10:54:48
35	1	1250	486.89	2010-08-18 10:55:48
36	1	1300	498.95	2010-08-18 10:56:38
37	1	1350	510.97	2010-08-18 10:57:38
38	1	1400	522.95	2010-08-18 10:58:38
39	1	1450	534.9	2010-08-18 10:59:28
40	1	1500	546.82	2010-08-18 11:00:28
41	1	1550	558.72	2010-08-18 11:01:18
42	1	1600	570.61	2010-08-18 11:02:18
43	1	1650	582.48	2010-08-18 11:03:18
44	1	1700	594.35	2010-08-18 11:04:08
45	1	1750	606.22	2010-08-18 11:05:08
46	1	1800	618.09	2010-08-18 11:05:58
47	1	1850	629.96	2010-08-18 11:06:58
48	1	1900	641.85	2010-08-18 11:07:58
49	1	1950	653.75	2010-08-18 11:08:48
50	1	2000	665.67	2010-08-18 11:09:48
51	1	2050	677.63	2010-08-18 11:10:48
52	1	2053.83	678.54	2010-08-18 11:11:18
53	1	2103.83	690.53	2010-08-18 11:12:18
54	1	2105.06	690.82	2010-08-18 12:36:18
55	1	2155.06	702.85	2010-08-18 12:37:18
56	1	2205.06	714.91	2010-08-18 12:38:18

57	1	2255.06	727.03	2010-08-18 12:39:18
58	1	2305.06	739.19	2010-08-18 12:40:18
59	1	2355.06	751.42	2010-08-18 12:41:18
60	1	2404.98	763.7	2010-08-18 12:42:18
61	1	2406.83	764.16	2010-08-18 12:50:28
62	1	2456.83	776.53	2010-08-18 12:51:38
63	1	2506.83	788.99	2010-08-18 12:53:08
64	1	2556.83	801.54	2010-08-18 12:54:08
65	1	2606.83	814.19	2010-08-18 12:54:58
66	1	2656.83	826.95	2010-08-18 12:56:08
67	1	2706.83	839.83	2010-08-18 12:57:08
68	1	2756.83	852.84	2010-08-18 12:58:08
69	1	2806.83	866	2010-08-18 12:59:08
70	1	2856.83	879.32	2010-08-18 13:00:08
71	1	2906.83	892.82	2010-08-18 13:01:38
72	1	2906.91	892.84	2010-08-18 13:02:18
73	1	2956.91	906.53	2010-08-18 13:03:28
74	1	3006.91	920.45	2010-08-18 13:04:28
75	1	3056.91	934.61	2010-08-18 13:05:28
76	1	3106.91	949.05	2010-08-18 13:06:18
77	1	3156.91	963.8	2010-08-18 13:07:28
78	1	3206.91	978.91	2010-08-18 13:08:18
79	1	3256.91	994.43	2010-08-18 13:09:18
80	1	3306.91	1010.43	2010-08-18 13:10:28
81	1	3356.91	1026.99	2010-08-18 13:11:28
82	1	3406.91	1044.25	2010-08-18 13:12:28
83	1	3456.91	1062.37	2010-08-18 13:13:28
84	1	3506.91	1081.59	2010-08-18 13:14:18
85	1	3556.91	1102.33	2010-08-18 13:15:18
86	1	3606.91	1125.32	2010-08-18 13:16:28
87	1	3656.91	1152.36	2010-08-18 13:17:28
88	1	3706.91	1193.49	2010-08-18 13:18:28

注：（1）罐体无变位进油，罐内油量初值 262L；
（2）第三列累加进油量是每次加入一定量油后的累加值；
（3）第四列是原罐内初始油量加入相应油量后油位高度值。

表 7-2 无变位出油

流水号	油罐号	累加出油量/L	油位高度/mm	采集时间
111	1	52.72	1150.72	2010-08-18 14:31:01
112	1	102.72	1123.99	2010-08-18 14:31:56
113	1	152.72	1101.15	2010-08-18 14:32:56
114	1	202.72	1080.51	2010-08-18 14:33:46
115	1	252.72	1061.36	2010-08-18 14:34:46

116	1	302.72	1043.29	2010-08-18 14:35:46
117	1	352.72	1026.08	2010-08-18 14:36:36
118	1	402.72	1009.54	2010-08-18 14:37:36
119	1	452.72	993.57	2010-08-18 14:38:26
120	1	502.72	978.08	2010-08-18 14:39:26
121	1	552.72	962.99	2010-08-18 14:40:16
122	1	602.72	948.26	2010-08-18 14:41:16
123	1	652.72	933.84	2010-08-18 14:42:06
124	1	702.72	919.69	2010-08-18 14:43:06
125	1	752.72	905.78	2010-08-18 14:43:56
126	1	802.72	892.1	2010-08-18 14:44:56
127	1	852.72	878.61	2010-08-18 14:45:46
128	1	902.72	865.3	2010-08-18 14:46:46
129	1	952.72	852.15	2010-08-18 14:47:36
130	1	1002.72	839.14	2010-08-18 14:48:36
131	1	1052.72	826.27	2010-08-18 14:49:26
132	1	1102.72	813.52	2010-08-18 14:50:26
133	1	1152.72	800.87	2010-08-18 14:51:16
134	1	1202.72	788.33	2010-08-18 14:52:16
135	1	1252.72	775.88	2010-08-18 14:53:16
136	1	1302.72	763.51	2010-08-18 14:54:06
137	1	1352.72	751.21	2010-08-18 14:55:06
138	1	1402.72	738.98	2010-08-18 14:56:06
139	1	1452.72	726.81	2010-08-18 14:56:56
140	1	1502.72	714.7	2010-08-18 14:57:56
141	1	1552.72	702.64	2010-08-18 14:58:46
142	1	1602.72	690.61	2010-08-18 14:59:46
143	1	1652.72	678.63	2010-08-18 15:00:36
144	1	1702.72	666.68	2010-08-18 15:01:36
145	1	1752.72	654.75	2010-08-18 15:02:36
146	1	1802.72	642.84	2010-08-18 15:03:26
147	1	1852.72	630.96	2010-08-18 15:04:26
148	1	1902.72	619.08	2010-08-18 15:05:16
149	1	1952.72	607.21	2010-08-18 15:06:16
150	1	2002.72	595.35	2010-08-18 15:07:06
151	1	2052.72	583.48	2010-08-18 15:08:06
152	1	2102.72	571.61	2010-08-18 15:09:06
153	1	2152.72	559.72	2010-08-18 15:09:56
154	1	2202.72	547.82	2010-08-18 15:10:56
155	1	2252.72	535.9	2010-08-18 15:11:46
156	1	2302.72	523.95	2010-08-18 15:12:46
157	1	2352.72	511.97	2010-08-18 15:13:36
158	1	2402.72	499.96	2010-08-18 15:14:36
159	1	2452.72	487.9	2010-08-18 15:15:36

160	1	2502.72	475.8	2010-08-18 15:16:26
161	1	2552.72	463.65	2010-08-18 15:17:26
162	1	2602.72	451.43	2010-08-18 15:18:26
163	1	2652.72	439.15	2010-08-18 15:19:16
164	1	2702.72	426.8	2010-08-18 15:20:16
165	1	2752.72	414.36	2010-08-18 15:21:16
166	1	2802.72	401.84	2010-08-18 15:22:06
167	1	2852.72	389.22	2010-08-18 15:23:06
168	1	2902.72	376.49	2010-08-18 15:23:56
169	1	2952.72	363.64	2010-08-18 15:24:56
170	1	3002.72	350.67	2010-08-18 15:25:56
171	1	3052.72	337.55	2010-08-18 15:26:46
172	1	3102.72	324.27	2010-08-18 15:27:46
173	1	3152.72	310.82	2010-08-18 15:28:46
174	1	3202.72	297.18	2010-08-18 15:29:36
175	1	3252.72	283.33	2010-08-18 15:30:36
176	1	3302.72	269.24	2010-08-18 15:31:26
177	1	3352.72	254.88	2010-08-18 15:32:26
178	1	3402.72	240.21	2010-08-18 15:33:26
179	1	3452.72	225.21	2010-08-18 15:34:26
180	1	3502.72	209.81	2010-08-18 15:35:16
181	1	3552.72	193.94	2010-08-18 15:36:16
182	1	3602.72	177.54	2010-08-18 15:37:16
183	1	3652.72	160.48	2010-08-18 15:38:06
184	1.00	3702.72	142.62	2010-08-18 15:39:06

注：（1）罐体无变位出油；
（2）第三列的累加出油量是每次抽出一定量油后的累加值；
（3）第四列是从罐内抽出相应油量后油位高度值。

表 7-3 倾斜变位进油

流水号	油罐号	累加进油量/L	油位高度/mm	采集时间
211	1	747.86	411.29	2010-08-19 15:10:27
212	1	797.86	423.45	2010-08-19 15:11:27
213	1	847.86	438.33	2010-08-19 15:12:37
214	1	897.86	450.54	2010-08-19 15:13:27
215	1	947.86	463.9	2010-08-19 15:14:27
216	1	997.86	477.74	2010-08-19 15:15:37
217	1	1047.86	489.37	2010-08-19 15:16:27
218	1	1097.79	502.56	2010-08-19 15:17:27
219	1	1147.79	514.69	2010-08-19 15:18:27
220	1	1197.73	526.84	2010-08-19 15:19:27
221	1	1247.73	538.88	2010-08-19 15:20:37
222	1	1297.73	551.96	2010-08-19 15:21:37

223	1	1347.73	564.4	2010-08-19 15:22:37
224	1	1397.73	576.56	2010-08-19 15:23:37
225	1	1447.73	588.74	2010-08-19 15:24:37
226	1	1497.73	599.56	2010-08-19 15:25:37
227	1	1547.73	611.62	2010-08-19 15:26:37
228	1	1597.73	623.44	2010-08-19 15:27:37
229	1	1647.73	635.58	2010-08-19 15:28:37
230	1	1697.73	646.28	2010-08-19 15:29:27
231	1	1747.73	658.59	2010-08-19 15:30:27
232	1	1797.73	670.22	2010-08-19 15:31:27
233	1	1847.73	680.63	2010-08-19 15:32:17
234	1	1897.73	693.03	2010-08-19 15:33:17
235	1	1947.73	704.67	2010-08-19 15:34:17
236	1	1997.73	716.45	2010-08-19 15:35:17
237	1	2047.73	727.66	2010-08-19 15:36:17
238	1	2097.73	739.39	2010-08-19 15:37:17
239	1	2147.73	750.9	2010-08-19 15:38:17
240	1	2197.73	761.55	2010-08-19 15:39:07
241	1	2247.73	773.43	2010-08-19 15:40:07
242	1	2297.73	785.39	2010-08-19 15:41:07
243	1	2347.73	796.04	2010-08-19 15:41:57
244	1	2397.73	808.27	2010-08-19 15:42:57
245	1	2447.73	820.8	2010-08-19 15:43:57
246	1	2497.73	832.8	2010-08-19 15:44:57
247	1	2547.73	844.47	2010-08-19 15:45:57
248	1	2597.73	856.29	2010-08-19 15:46:57
249	1	2647.73	867.6	2010-08-19 15:47:47
250	1	2697.73	880.06	2010-08-19 15:48:47
251	1	2747.73	892.92	2010-08-19 15:49:57
252	1	2797.73	904.34	2010-08-19 15:50:57
253	1	2847.73	917.34	2010-08-19 15:51:57
254	1	2897.73	929.9	2010-08-19 15:52:57
255	1	2947.73	941.42	2010-08-19 15:53:47
256	1	2997.73	954.6	2010-08-19 15:54:47
257	1	3047.73	968.09	2010-08-19 15:55:47
258	1	3097.73	980.14	2010-08-19 15:56:37
259	1	3147.73	992.41	2010-08-19 15:57:37
260	1	3197.73	1006.34	2010-08-19 15:58:37
261	1	3247.73	1019.07	2010-08-19 15:59:27
262	1	3297.73	1034.24	2010-08-19 16:00:37
263	1	3299.74	1035.36	2010-08-19 16:01:27

注：（1）罐体纵向倾斜变位进油，纵向倾斜角 4.1° ，罐内油量初值 215L；

（2）第三列累加进油量是每次加入一定量油后的累加值；

（3）第四列是原罐内油量加入相应油量后油位高度值。

表 7-4 倾斜变位出油

流水号	油罐号	累加出油量/L	油位高度/mm	采集时间
311	1	50	1020.65	2010-08-19 16:10:31
312	1	100	1007.73	2010-08-19 16:11:21
313	1	150	994.32	2010-08-19 16:12:21
314	1	200	980.96	2010-08-19 16:13:11
315	1	250	967.1	2010-08-19 16:14:11
316	1	300	956.01	2010-08-19 16:15:01
317	1	350	941.54	2010-08-19 16:16:01
318	1	400	929.69	2010-08-19 16:16:51
319	1	450	916.44	2010-08-19 16:17:51
320	1	500	904.14	2010-08-19 16:18:51
321	1	550	891.9	2010-08-19 16:19:41
322	1	600	879.23	2010-08-19 16:20:41
323	1	650	868.99	2010-08-19 16:21:31
324	1	700	855.13	2010-08-19 16:22:31
325	1	750	844.02	2010-08-19 16:23:21
326	1	800	831.64	2010-08-19 16:24:21
327	1	850	820.47	2010-08-19 16:25:11
328	1	900	808.16	2010-08-19 16:26:11
329	1	950	796	2010-08-19 16:27:11
330	1	1000	785.04	2010-08-19 16:28:01
331	1	1050	773.07	2010-08-19 16:29:01
332	1	1100	762.09	2010-08-19 16:29:51
333	1	1150	750.81	2010-08-19 16:30:51
334	1	1200	739.42	2010-08-19 16:31:41
335	1	1250	727.09	2010-08-19 16:32:41
336	1	1300	715.32	2010-08-19 16:33:41
337	1	1350	705.43	2010-08-19 16:34:31
338	1	1400	693.52	2010-08-19 16:35:31
339	1	1450	682.5	2010-08-19 16:36:21
340	1	1500	671.02	2010-08-19 16:37:21
341	1	1550	658.68	2010-08-19 16:38:21
342	1	1600	647.74	2010-08-19 16:39:11
343	1	1650	635.76	2010-08-19 16:40:11
344	1	1700	624.61	2010-08-19 16:41:01
345	1	1750	612.53	2010-08-19 16:42:01
346	1	1800	600.69	2010-08-19 16:43:01
347	1	1850	589.4	2010-08-19 16:43:51
348	1	1900	577	2010-08-19 16:44:51
349	1	1950	564.58	2010-08-19 16:45:51
350	1	2000	554.33	2010-08-19 16:46:41
351	1	2050	540.76	2010-08-19 16:47:41

352	1	2100	528.65	2010-08-19 16:48:41
353	1	2150	517.19	2010-08-19 16:49:31
354	1	2200	504.87	2010-08-19 16:50:31
355	1	2250	490.78	2010-08-19 16:51:51
356	1	2300	478.06	2010-08-19 16:52:51
357	1	2350	465.97	2010-08-19 16:53:41
358	1	2400	452.4	2010-08-19 16:54:41
359	1	2450	439.98	2010-08-19 16:55:31
360	1	2500	425.83	2010-08-19 16:56:31
361	1	2550	411.73	2010-08-19 16:57:31

注：（1）罐体纵向倾斜变位出油，纵向倾斜角 4.1° ；
（2）第三列累加出油量是每次抽出一定量油后的累加值；
（3）第四列是原罐内油量抽出一定量油后的油位高度值。

§ 7.2 附件 2

表 7-5 实际储油罐的采集数据

流水号	油罐号	进油量 /L	出油量/L	显示油高/mm	显示油量容积 /L	采集时间
201	02	0	60.00	2632.23	60448.88	2010-08-01 08:00:49
202	02	0	149.09	2624.30	60311.43	2010-08-01 08:15:42
203	02	0	68.45	2620.67	60248.03	2010-08-01 08:23:41
204	02	0	199.27	2610.29	60065.11	2010-08-01 08:38:14
205	02	0	70.05	2606.61	59999.69	2010-08-01 08:53:08
206	02	0	136.36	2599.59	59874.06	2010-08-01 09:08:01
207	02	0	232.74	2587.60	59657.02	2010-08-01 09:32:06
208	02	0	107.97	2582.05	59555.51	2010-08-01 09:56:18
209	02	0	49.24	2579.57	59509.94	2010-08-01 10:10:09
210	02	0	80.65	2575.44	59433.77	2010-08-01 10:26:28
211	02	0	120.29	2569.46	59322.85	2010-08-01 10:48:35
212	02	0	108.24	2564.12	59223.17	2010-08-01 11:02:48
213	02	0	83.46	2559.83	59142.66	2010-08-01

						11:36:04
214	02	0	229.93	2548.47	58927.69	2010-08-01 13:16:24
215	02	0	181.70	2539.63	58758.61	2010-08-01 13:24:25
216	02	0	238.52	2528.01	58534.01	2010-08-01 13:37:33
217	02	0	131.79	2521.63	58409.58	2010-08-01 14:00:03
218	02	0	238.33	2510.23	58185.31	2010-08-01 14:36:02
219	02	0	42.92	2508.17	58144.52	2010-08-01 14:52:08
220	02	0	171.34	2500.07	57983.36	2010-08-01 15:06:14
221	02	0	212.34	2490.06	57782.53	2010-08-01 15:16:43
222	02	0	92.38	2485.73	57695.08	2010-08-01 15:48:22
223	02	0	243.85	2474.40	57464.67	2010-08-01 16:06:31
224	02	0	206.69	2464.77	57267.02	2010-08-01 16:25:54
225	02	0	224.50	2454.51	57054.65	2010-08-01 16:41:24
226	02	0	169.26	2446.77	56893.24	2010-08-01 17:06:08
227	02	0	220.09	2436.85	56684.86	2010-08-01 17:45:54
228	02	0	117.54	2431.55	56572.86	2010-08-01 18:03:34
229	02	0	93.44	2427.32	56483.12	2010-08-01 18:14:52
230	02	0	114.46	2422.20	56374.11	2010-08-01 18:35:24
231	02	0	174.69	2414.35	56206.14	2010-08-01 19:12:55
232	02	0	232.77	2404.05	55984.22	2010-08-01 20:34:44
233	02	0	110.86	2399.15	55878.05	2010-08-01 21:05:57
234	02	0	138.59	2393.12	55746.87	2010-08-02 07:18:34
235	02	0	242.21	2382.50	55514.45	2010-08-02

						07:23:30
236	02	0	186.43	2374.35	55334.90	2010-08-02
						07:39:21
237	02	0	275.38	2362.44	55070.68	2010-08-02
						07:48:36
238	02	0	92.65	2358.40	54980.57	2010-08-02
						08:04:44
239	02	0	239.28	2348.13	54750.40	2010-08-02
						08:28:32
240	02	0	206.68	2339.37	54552.85	2010-08-02
						08:33:36
241	02	0	104.63	2334.88	54451.17	2010-08-02
						08:46:28
242	02	0	158.80	2328.13	54297.75	2010-08-02
						08:58:39
243	02	0	142.43	2322.14	54161.06	2010-08-02
						09:03:54
244	02	0	189.17	2314.14	53977.71	2010-08-02
						09:14:47
245	02	0	238.95	2304.14	53747.27	2010-08-02
						09:25:55
246	02	0	73.58	2301.09	53676.72	2010-08-02
						09:35:54
247	02	0	245.27	2290.87	53439.37	2010-08-02
						09:59:33
248	02	0	251.78	2280.46	53196.16	2010-08-02
						10:13:32
249	02	0	134.59	2274.92	53066.14	2010-08-02
						10:18:24
250	02	0	153.02	2268.61	52917.56	2010-08-02
						10:26:36
251	02	0	188.26	2260.89	52735.08	2010-08-02
						10:38:33
252	02	0	220.97	2251.88	52521.14	2010-08-02
						10:46:54
253	02	0	229.97	2242.46	52296.37	2010-08-02
						10:58:37
254	02	0	237.73	2232.88	52066.65	2010-08-02
						11:05:34
255	02	0	144.04	2226.99	51924.85	2010-08-02
						11:26:38
256	02	0	158.25	2220.70	51772.96	2010-08-02
						11:58:37
257	02	0	287.87	2209.13	51492.32	2010-08-02

						12:09:46
258	02	0	192.77	2201.40	51303.95	2010-08-02 12:36:44
259	02	0	262.67	2190.91	51047.20	2010-08-02 12:55:32
260	02	0	121.96	2186.14	50930.04	2010-08-02 13:18:34
261	02	0	208.47	2177.92	50727.53	2010-08-02 13:26:42
262	02	0	198.58	2170.04	50532.69	2010-08-02 13:38:24
263	02	0	297.29	2158.40	50243.62	2010-08-02 13:45:37
264	02	0	72.41	2155.54	50172.37	2010-08-02 13:55:34
265	02	0	178.40	2148.54	49997.62	2010-08-02 14:18:36
266	02	0	184.06	2141.32	49816.82	2010-08-02 14:27:33
267	02	0	74.38	2138.42	49744.05	2010-08-02 14:48:24
268	02	0	285.23	2127.37	49465.96	2010-08-02 14:56:38
269	02	0	279.15	2116.53	49191.94	2010-08-02 15:05:14
270	02	0	166.23	2110.14	49029.86	2010-08-02 15:16:38
271	02	0	254.41	2100.32	48779.99	2010-08-02 15:27:54
272	02	0	89.64	2096.84	48691.21	2010-08-02 15:48:34
273	02	0	214.75	2088.64	48481.56	2010-08-02 15:56:39
274	02	0	120.77	2084.03	48363.42	2010-08-02 16:23:44
275	02	0	168.81	2077.58	48197.79	2010-08-02 16:32:36
276	02	0	272.92	2067.14	47928.87	2010-08-02 16:54:32
277	02	0	103.92	2063.17	47826.34	2010-08-02 17:38:34
278	02	0	131.94	2058.14	47696.24	2010-08-02 17:45:26
279	02	0	181.74	2051.30	47518.96	2010-08-02

						18:08:37
280	02	0	142.11	2045.92	47379.23	2010-08-02
						19:24:24
281	02	0	264.62	2035.94	47119.35	2010-08-02
						19:48:54
282	02	0	313.84	2024.06	46808.88	2010-08-02
						20:08:36
283	02	0	96.66	2020.47	46714.83	2010-08-02
						20:13:24
284	02	0	116.16	2016.11	46600.46	2010-08-02
						20:43:34
285	02	0	239.12	2007.08	46363.09	2010-08-03
						07:01:55
286	02	0	154.50	2001.33	46211.59	2010-08-03
						07:47:25
287	02	0	314.56	1989.59	45901.47	2010-08-03
						08:03:51
288	02	0	316.03	1977.87	45590.80	2010-08-03
						08:15:35
289	02	0	226.29	1969.43	45366.43	2010-08-03
						08:33:26
290	02	0	285.90	1958.83	45083.88	2010-08-03
						08:53:45
291	02	0	163.75	1952.81	44923.05	2010-08-03
						09:04:33
292	02	0	224.69	1944.49	44700.35	2010-08-03
						09:27:36
293	02	0	321.69	1932.64	44382.33	2010-08-03
						10:02:12
294	02	0	205.69	1925.05	44178.12	2010-08-03
						10:57:23
295	02	0	309.66	1913.71	43872.31	2010-08-03
						11:22:45
296	02	0	249.73	1904.51	43623.60	2010-08-03
						11:35:36
297	02	0	186.43	1897.67	43438.33	2010-08-03
						12:31:53
298	02	0	231.42	1889.27	43210.41	2010-08-03
						13:37:51
299	02	0	297.79	1878.40	42914.84	2010-08-03
						13:57:25
300	02	0	109.19	1874.41	42806.16	2010-08-03
						14:21:51
301	02	0	162.87	1868.46	42643.93	2010-08-03

						14:36:23
302	02	0	328.52	1856.54	42318.32	2010-08-03
						14:47:55
303	02	0	166.13	1850.51	42153.30	2010-08-03
						15:14:34
304	02	0	237.66	1841.92	41917.87	2010-08-03
						15:34:26
305	02	0	303.97	1830.91	41615.55	2010-08-03
						15:58:50
306	02	0	330.11	1818.96	41286.70	2010-08-03
						16:17:44
307	02	0	235.69	1810.42	41051.26	2010-08-03
						17:37:12
308	02	0	86.15	1807.34	40966.26	2010-08-03
						17:44:56
309	02	0	66.88	1804.98	40901.09	2010-08-03
						18:07:55
310	02	0	225.69	1796.80	40675.04	2010-08-03
						18:14:46
311	02	0	213.49	1789.13	40462.79	2010-08-03
						18:35:43
312	02	0	323.39	1777.51	40140.74	2010-08-03
						19:07:24
313	02	0	263.59	1768.05	39878.12	2010-08-03
						20:42:51
314	02	0	240.84	1759.42	39638.21	2010-08-04
						08:21:22
315	02	0	201.31	1752.19	39437.00	2010-08-04
						08:32:26
316	02	0	128.80	1747.59	39308.87	2010-08-04
						08:41:46
317	02	0	324.38	1735.97	38984.85	2010-08-04
						08:53:32
318	02	0	206.56	1728.57	38778.25	2010-08-04
						09:21:22
319	02	0	64.47	1726.26	38713.71	2010-08-04
						09:55:27
320	02	0	251.13	1717.30	38463.22	2010-08-04
						10:11:28
321	02	0	201.55	1710.10	38261.74	2010-08-04
						10:34:22
322	02	0	72.35	1707.51	38189.22	2010-08-04
						10:51:26
323	02	0	305.18	1696.61	37883.79	2010-08-04

						11:15:52
324	02	0	148.15	1691.31	37735.15	2010-08-04 11:31:55
325	02	0	120.83	1687.01	37614.49	2010-08-04 11:45:23
326	02	0	88.98	1683.87	37526.35	2010-08-04 12:01:22
327	02	0	142.33	1678.77	37383.13	2010-08-04 12:17:27
328	02	0	121.05	1674.48	37262.60	2010-08-04 12:51:12
329	02	0	240.40	1665.96	37023.08	2010-08-04 13:53:26
330	02	0	75.05	1663.30	36948.26	2010-08-04 14:11:25
331	02	0	133.08	1658.55	36814.61	2010-08-04 15:01:42
332	02	0	134.22	1653.73	36678.94	2010-08-04 15:25:28
333	02	0	303.81	1642.93	36374.74	2010-08-04 16:44:42
334	02	0	181.64	1636.48	36192.94	2010-08-04 17:41:33
335	02	0	268.23	1626.92	35923.33	2010-08-04 18:15:44
336	02	0	226.45	1618.89	35696.73	2010-08-04 18:31:22
337	02	0	276.49	1609.06	35419.17	2010-08-04 19:13:26
338	02	0	87.44	1605.92	35330.47	2010-08-04 19:31:45
339	02	0	331.77	1594.13	34997.30	2010-08-04 19:44:23
340	02	0	293.80	1583.65	34700.98	2010-08-04 19:56:42
341	02	0	72.05	1581.14	34629.99	2010-08-04 20:31:21
342	02	0	187.88	1574.48	34441.58	2010-08-05 06:55:34
343	02	0	148.35	1569.22	34292.75	2010-08-05 07:26:32
344	02	0	233.46	1560.92	34057.83	2010-08-05 07:35:14
345	02	0	120.98	1556.62	33936.10	2010-08-05

						07:50:32
346	02	0	220.76	1548.82	33715.25	2010-08-05 08:05:34
347	02	0	224.72	1540.79	33487.84	2010-08-05 08:18:32
348	02	0	224.46	1532.79	33261.23	2010-08-05 08:30:56
349	02	0	183.46	1526.30	33077.38	2010-08-05 08:44:24
350	02	0	65.53	1523.95	33010.80	2010-08-05 08:55:34
351	02	0	200.46	1516.81	32808.51	2010-08-05 09:06:27
352	02	0	170.84	1510.73	32636.23	2010-08-05 09:17:34
353	02	0	86.76	1507.65	32548.96	2010-08-05 09:28:28
354	02	0	187.61	1501.06	32362.23	2010-08-05 09:40:39
355	02	0	181.73	1494.55	32177.83	2010-08-05 09:56:24
356	02	0	210.27	1487.03	31964.75	2010-08-05 10:15:56
357	02	0	282.54	1476.98	31680.00	2010-08-05 10:25:32
358	02	0	253.04	1467.97	31424.74	2010-08-05 10:42:14
359	02	0	300.69	1457.25	31121.10	2010-08-05 10:55:54
360	02	0	70.11	1454.73	31049.73	2010-08-05 11:06:30
361	02	0	118.47	1450.53	30930.79	2010-08-05 11:45:24
362	02	0	185.56	1443.93	30743.92	2010-08-05 12:03:31
363	02	0	323.97	1432.35	30416.13	2010-08-05 12:15:34
364	02	0	277.01	1422.45	30136.01	2010-08-05 12:26:35
365	02	0	181.51	1415.93	29951.59	2010-08-05 12:48:22
366	02	0	149.46	1410.60	29800.87	2010-08-05 13:15:37
367	02	0	72.20	1408.01	29727.64	2010-08-05

						13:36:24
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611	02	0	64.41	1755.36	39525.25	2010-08-10 13:25:12
612	02	0	74.82	1752.67	39450.36	2010-08-10 13:36:42
613	02	0	145.22	1747.47	39305.53	2010-08-10 13:45:54
614	02	0	204.36	1740.16	39101.75	2010-08-10 13:57:33
615	02	0	239.00	1731.62	38863.43	2010-08-10 14:08:26
616	02	0	170.27	1725.54	38693.59	2010-08-10 14:22:12
617	02	0	284.73	1715.34	38408.39	2010-08-10 14:35:55
618	02	0	257.72	1706.16	38151.41	2010-08-10 14:46:16
619	02	0	327.08	1694.47	37823.78	2010-08-10 14:57:48
620	02	0	204.55	1687.16	37618.70	2010-08-10 15:06:42
621	02	0	149.65	1681.91	37471.31	2010-08-10 15:15:12
622	02	0	83.02	1678.85	37385.37	2010-08-10 15:27:51
623	02	0	229.38	1670.74	37157.48	2010-08-10 15:35:12
624	02	0	273.31	1660.96	36882.43	2010-08-10 15:46:36
625	02	0	175.17	1654.72	36706.81	2010-08-10 15:55:13
626	02	0	83.88	1651.81	36624.88	2010-08-10 16:07:32
627	02	0	129.82	1647.15	36493.63	2010-08-10 16:15:12
628	02	0	97.24	1643.61	36393.90	2010-08-10 16:28:52
629	02	0	137.82	1638.80	36258.34	2010-08-10 16:35:12
630	02	0	179.47	1632.40	36077.90	2010-08-10 16:46:17
631	02	0	204.59	1625.13	35872.83	2010-08-10

						16:55:12
632	02	0	184.45	1618.55	35687.13	2010-08-10 17:08:18
633	02	0	302.62	1607.80	35383.58	2010-08-10 17:15:52
634	02	0	203.48	1600.62	35180.73	2010-08-10 17:24:15
635	02	0	321.65	1589.18	34857.36	2010-08-10 17:35:12
636	02	0	235.05	1580.81	34620.66	2010-08-10 17:46:14
637	02	0	326.24	1569.23	34293.03	2010-08-10 17:55:12
638	02	0	122.32	1564.82	34168.22	2010-08-10 18:06:15
639	02	0	248.05	1556.06	33920.25	2010-08-10 18:25:12
640	02	0	137.65	1551.17	33781.79	2010-08-10 18:35:00
641	02	0	246.30	1542.45	33534.85	2010-08-10 19:00:10
642	02	0	252.17	1533.50	33281.35	2010-08-10 19:15:12
643	02	0	73.94	1530.82	33205.43	2010-08-10 19:45:45
644	02	0	128.71	1526.27	33076.53	2010-08-10 19:55:24
645	02	0	118.68	1522.03	32956.40	2010-08-10 20:06:16
646	02	0	244.88	1513.36	32710.75	2010-08-10 20:35:12
647	02	0	292.62	1502.91	32414.65	2010-08-10 20:51:35
648	02	0	154.80	1497.47	32260.57	2010-08-11 08:01:49
649	02	0	273.28	1487.66	31982.60	2010-08-11 08:14:41
650	02	0	246.25	1478.91	31734.68	2010-08-11 08:23:44
651	02	0	58.64	1476.84	31676.03	2010-08-11 08:38:16
652	02	0	226.02	1468.82	31448.82	2010-08-11 08:46:28
653	02	0	164.19	1462.95	31282.55	2010-08-11

						08:55:01
654	02	0	312.54	1451.79	30966.47	2010-08-11 09:10:34
655	02	0	55.66	1449.78	30909.56	2010-08-11 09:26:28
656	02	0	187.10	1443.16	30722.12	2010-08-11 09:35:09
657	02	0	173.45	1436.91	30545.19	2010-08-11 09:56:28
658	02	0	184.05	1430.30	30358.12	2010-08-11 10:04:35
659	02	0	273.68	1420.60	30083.68	2010-08-11 11:00:48
660	02	0	145.31	1415.38	29936.03	2010-08-11 11:36:04
661	02	0	276.05	1405.53	29657.54	2010-08-11 13:16:26
662	02	0	188.11	1398.82	29467.90	2010-08-11 13:24:25
663	02	0	63.63	1396.46	29401.21	2010-08-11 13:37:33
664	02	0	105.47	1392.70	29294.99	2010-08-11 14:03:03
665	02	0	257.94	1383.48	29034.63	2010-08-11 14:36:02
666	02	0	187.06	1376.75	28844.67	2010-08-11 14:52:08
667	02	0	98.55	1373.22	28745.07	2010-08-11 15:04:16
668	02	0	151.44	1367.81	28592.47	2010-08-11 15:16:43
669	02	0	225.70	1359.73	28364.67	2010-08-11 15:38:22
670	02	0	106.68	1355.82	28254.49	2010-08-11 15:44:31
671	02	0	261.04	1346.43	27990.02	2010-08-11 15:55:24
672	02	0	122.99	1342.00	27865.32	2010-08-11 16:11:24
673	02	0	309.81	1330.83	27551.11	2010-08-11 16:32:08
674	02	0	130.53	1326.14	27419.28	2010-08-11 17:45:54
675	02	0	266.27	1316.48	27147.93	2010-08-11

						18:03:34
676	02	0	106.76	1312.60	27039.01	2010-08-11 18:14:55
677	02	0	135.63	1307.72	26902.09	2010-08-11 18:35:24
678	02	0	79.26	1304.81	26820.47	2010-08-11 19:12:55
679	02	0	214.49	1297.05	26602.96	2010-08-11 20:34:44
680	02	0	243.09	1288.22	26355.68	2010-08-11 21:05:57
681	02	0	206.41	1280.75	26146.69	2010-08-12 07:18:34
682	02	0	171.54	1274.49	25971.69	2010-08-12 07:23:22
683	02	0	230.76	1266.05	25735.97	2010-08-12 07:39:21
684	02	0	233.21	1257.57	25499.40	2010-08-12 07:48:36
685	02	0	239.81	1248.78	25254.46	2010-08-12 08:04:43
686	02	0	227.09	1240.42	25021.78	2010-08-12 08:25:32
687	02	0	313.02	1228.97	24703.56	2010-08-12 08:33:36
688	02	0	111.18	1224.88	24590.03	2010-08-12 08:46:28
689	02	0	246.20	1215.79	24337.95	2010-08-12 08:58:39
690	02	0	119.20	1211.43	24217.17	2010-08-12 09:03:54
691	02	0	87.35	1208.23	24128.58	2010-08-12 09:14:47
692	02	0	217.58	1200.16	23905.36	2010-08-12 09:25:51
693	02	0	177.85	1193.66	23725.80	2010-08-12 09:35:54
694	02	0	177.88	1187.07	23543.96	2010-08-12 09:59:33
695	02	0	230.61	1178.45	23306.41	2010-08-12 10:13:32
696	02	0	262.13	1168.75	23039.56	2010-08-12 10:18:24
697	02	0	148.12	1163.25	22888.46	2010-08-12

						10:24:36
698	02	0	230.85	1154.63	22651.98	2010-08-12
						10:38:33
699	02	0	165.42	1148.47	22483.22	2010-08-12
						10:46:54
700	02	0	278.40	1138.05	22198.25	2010-08-12
						10:58:37
701	02	0	275.32	1127.72	21916.34	2010-08-12
						11:05:36
702	02	0	121.64	1123.15	21791.82	2010-08-12
						11:26:38
703	02	0	216.12	1115.02	21570.61	2010-08-12
						11:58:37
704	02	0	206.56	1107.20	21358.20	2010-08-12
						12:09:46
705	02	0	196.22	1099.79	21157.29	2010-08-12
						12:36:44
706	02	0	280.73	1089.09	20867.78	2010-08-12
						12:55:32
707	02	0	123.17	1084.44	20742.19	2010-08-12
						13:16:35
708	02	0	135.54	1079.26	20602.45	2010-08-12
						13:26:42
709	02	0	84.51	1076.07	20516.49	2010-08-12
						13:38:24
710	02	0	298.42	1064.67	20209.84	2010-08-12
						13:45:37
711	02	0	219.88	1056.21	19982.86	2010-08-12
						13:56:35
712	02	0	176.67	1049.42	19801.04	2010-08-12
						14:18:36
713	02	0	218.93	1041.03	19576.83	2010-08-12
						14:27:33
714	02	0	191.84	1033.58	19378.17	2010-08-12
						14:48:24
715	02	0	219.72	1025.11	19152.81	2010-08-12
						14:56:38
716	02	0	191.82	1017.67	18955.30	2010-08-12
						15:05:14
717	02	0	235.36	1008.46	18711.39	2010-08-12
						15:16:38
718	02	0	185.72	1001.23	18520.39	2010-08-12
						15:27:54
719	02	0	305.06	989.30	18206.12	2010-08-12

						15:48:34
720	02	0	107.03	985.11	18096.02	2010-08-12 15:56:39
721	02	0	78.44	982.05	18015.71	2010-08-12 16:23:44
722	02	0	78.44	978.95	17934.42	2010-08-12 16:32:36
723	02	0	65.34	976.32	17865.52	2010-08-12 16:54:32
724	02	0	155.92	970.27	17707.24	2010-08-12 17:38:34
725	02	0	162.75	963.79	17538.06	2010-08-12 17:45:26
726	02	0	142.36	958.13	17390.58	2010-08-12 18:08:37
727	02	0	241.74	948.50	17140.30	2010-08-12 19:24:24
728	02	0	208.69	940.22	16925.76	2010-08-12 19:48:54
729	02	0	240.84	930.50	16674.70	2010-08-12 20:08:36
730	02	0	281.26	919.17	16383.16	2010-08-12 20:13:24
731	02	0	291.06	907.44	16082.58	2010-08-12 20:43:34
732	02	0	96.06	903.52	15982.43	2010-08-13 07:01:55
733	02	0	84.03	900.13	15895.94	2010-08-13 07:47:25
734	02	0	217.70	891.17	15667.87	2010-08-13 08:03:51
735	02	0	72.77	888.23	15593.21	2010-08-13 08:15:35
736	02	0	177.34	880.98	15409.46	2010-08-13 08:33:26
737	02	0	178.20	873.67	15224.73	2010-08-13 08:53:45
738	02	0	256.15	863.06	14957.57	2010-08-13 09:04:33
739	02	0	164.02	856.21	14785.71	2010-08-13 09:27:36
740	02	0	143.51	850.28	14637.34	2010-08-13 10:02:12
741	02	0	208.74	841.56	14419.83	2010-08-13

						10:57:23
742	02	0	225.36	832.15	14186.04	2010-08-13 11:22:45
743	02	0	169.90	824.95	14007.80	2010-08-13 11:35:36
744	02	0	129.18	819.47	13872.54	2010-08-13 12:31:53
745	02	0	84.11	815.97	13786.32	2010-08-13 13:37:51
746	02	0	183.13	808.11	13593.21	2010-08-13 13:57:25
747	02	0	107.36	803.49	13480.03	2010-08-13 14:21:51
748	02	0	59.93	801.05	13420.35	2010-08-13 14:36:23
749	02	0	223.06	791.50	13187.45	2010-08-13 14:47:55
750	02	0	101.51	787.07	13079.78	2010-08-13 15:14:34
751	02	0	148.07	780.65	12924.16	2010-08-13 15:34:26
752	02	0	205.31	771.77	12709.71	2010-08-13 15:58:50
753	02	0	128.35	766.18	12575.21	2010-08-13 16:17:44
754	02	0	214.02	756.81	12350.62	2010-08-13 17:37:12
755	02	0	135.84	750.87	12208.81	2010-08-13 17:44:56
756	02	0	200.57	742.03	11998.59	2010-08-13 18:07:55
757	02	0	201.28	732.99	11784.64	2010-08-13 18:14:46
758	02	0	144.71	726.57	11633.34	2010-08-13 18:35:43
759	02	0	50.19	724.37	11581.62	2010-08-13 19:07:24
760	02	0	116.75	719.06	11457.04	2010-08-13 20:42:51
761	02	0	140.18	712.82	11311.13	2010-08-14 08:21:22
762	02	0	103.13	708.12	11201.56	2010-08-14 08:32:26
763	02	0	86.97	704.15	11109.25	2010-08-14

						08:41:46
764	02	0	225.39	693.93	10872.59	2010-08-14 08:53:32
765	02	0	137.10	687.63	10727.42	2010-08-14 09:21:22
766	02	0	235.68	676.75	10478.02	2010-08-14 09:55:27
767	02	0	126.01	670.84	10343.25	2010-08-14 10:11:28
768	02	0	208.71	661.15	10123.36	2010-08-14 10:34:22
769	02	0	127.78	655.18	9988.56	2010-08-14 10:51:26
770	02	0	213.28	645.10	9762.14	2010-08-14 11:15:52
771	02	0	200.68	635.55	9549.03	2010-08-14 11:31:55
772	02	0	119.58	629.77	9420.71	2010-08-14 11:45:23
773	02	0	87.63	625.61	9328.67	2010-08-14 12:01:22
774	02	0	205.05	615.71	9110.70	2010-08-14 12:17:27
775	02	0	234.92	604.21	8859.42	2010-08-14 12:51:12
776	02	0	107.45	598.94	8744.97	2010-08-14 13:53:26
777	02	0	175.48	590.23	8556.78	2010-08-14 14:11:25
778	02	0	128.20	583.84	8419.49	2010-08-14 15:03:41
779	02	0	206.00	573.50	8198.76	2010-08-14 15:25:28
780	02	0	191.45	563.82	7993.72	2010-08-14 16:44:42
781	02	0	72.20	560.14	7916.18	2010-08-14 17:41:33
782	02	0	206.04	549.52	7693.71	2010-08-14 18:15:44
783	02	0	231.25	537.63	7446.94	2010-08-14 18:31:22
784	02	0	135.86	530.48	7299.73	2010-08-14 19:13:26
785	02	0	205.87	519.64	7078.29	2010-08-14

						19:31:34
786	02	0	145.64	511.76	6918.64	2010-08-14 19:43:23
787	02	0	66.65	508.21	6847.08	2010-08-14 19:56:42
788	02	0	74.31	504.21	6766.73	2010-08-14 20:31:21
789	02	0	112.80	498.14	6645.37	2010-08-15 06:55:34
790	02	0	171.72	488.66	6457.19	2010-08-15 07:26:32
791	02	0	185.46	478.40	6255.45	2010-08-15 07:35:14
792	02	0	177.26	468.50	6062.70	2010-08-15 08:04:32
793	02	0	91.52	463.31	5962.42	2010-08-15 08:15:34
794	02	0	128.34	455.97	5821.49	2010-08-15 08:22:32
795	02	0	50.13	453.07	5766.11	2010-08-15 08:35:56
796	02	0	54.91	449.96	5706.90	2010-08-15 08:47:24
797	02	0	57.17	446.59	5642.96	2010-08-15 08:55:34
798	02	0	149.83	437.81	5477.45	2010-08-15 09:06:27
799	02	0	118.46	430.86	5347.56	2010-08-15 09:15:34
800	02	0	65.81	426.96	5275.11	2010-08-15 09:23:28
801	02	0	115.30	420.01	5146.78	2010-08-15 09:35:39
802	02	0	57.09	416.53	5082.90	2010-08-15 09:46:24
803	02	0	43.13	413.98	5036.26	2010-08-15 09:55:56

注：（1）第三列进油量；
（2）第四列出油量是加油机不同时段出的油量 v_i ；
（3）第五列显示油高是进/出油 v_i 后对应的罐内，油位显示高度 h_i ；
（4）第六列显示油量容积是进/出油 v_i 后，罐内，油位显示高度为 h_i 时系统的显示储油量 V_i 。

§ 7.3 图形

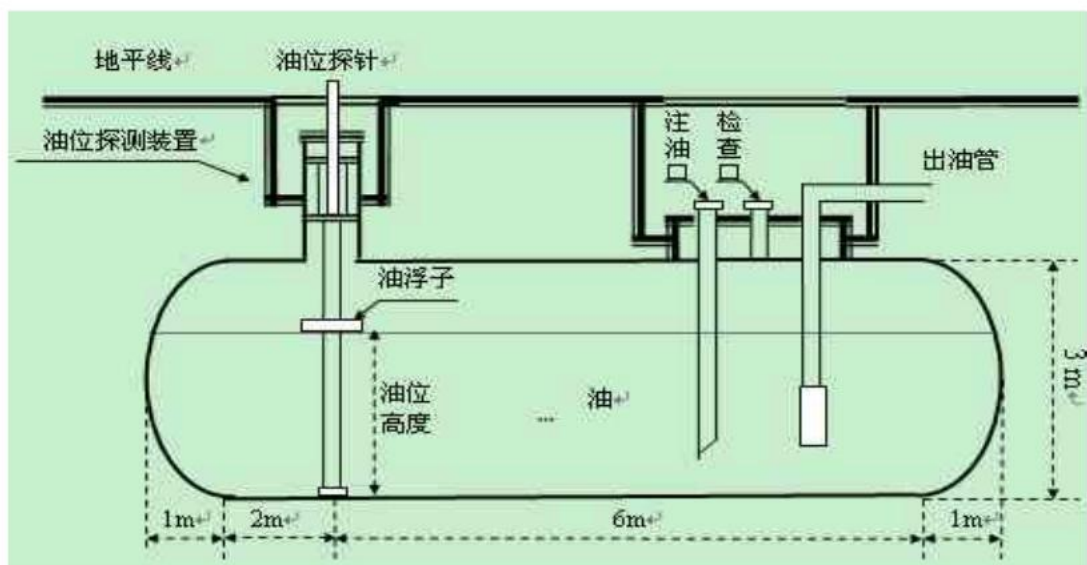


图 1 储油罐正面示意图

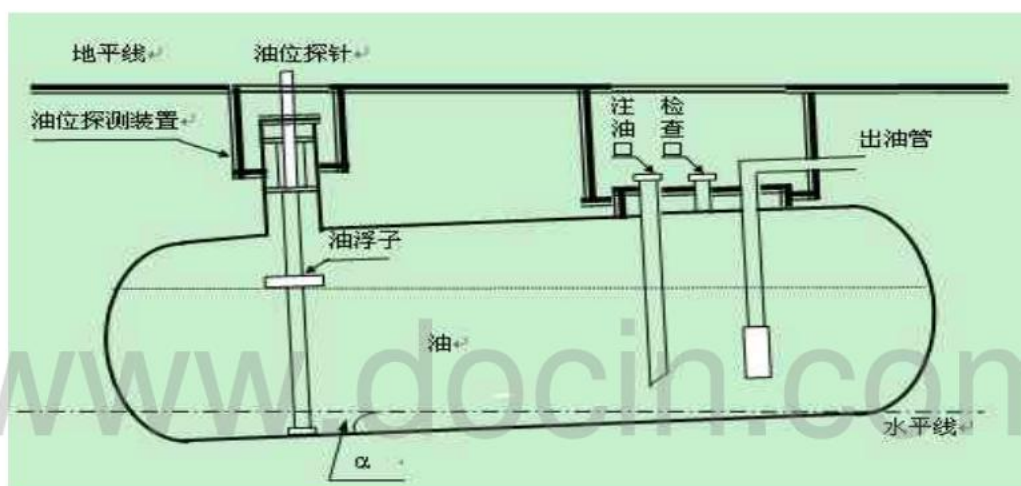


图 2 储油罐纵向倾斜变位后示意

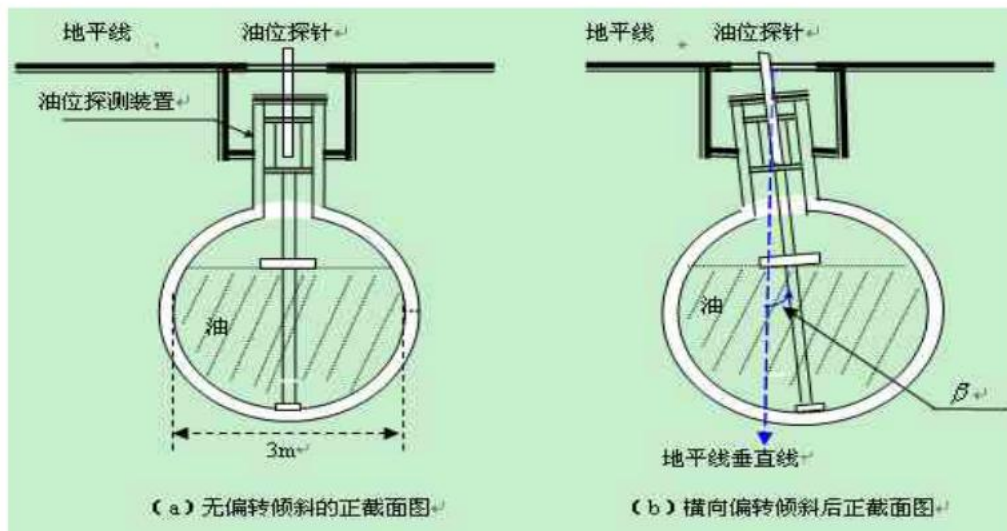


图 3 储油罐截面示意图

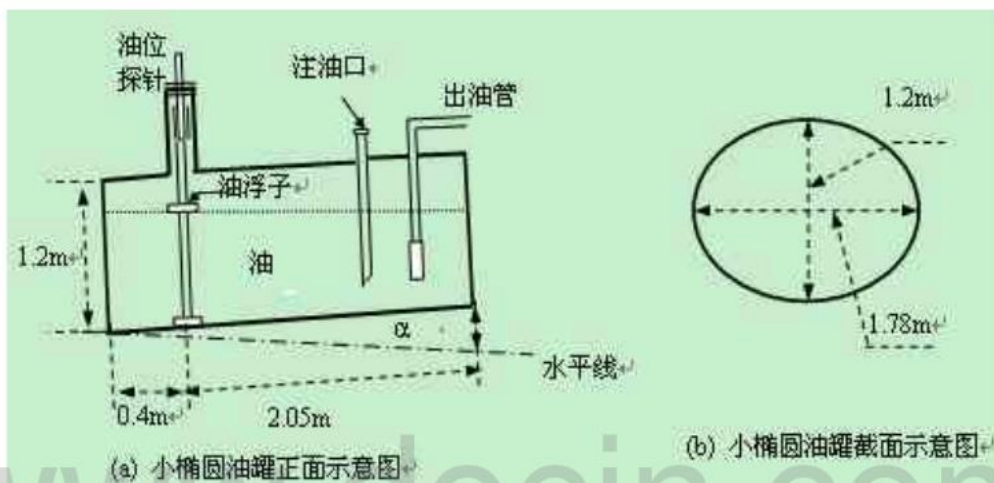


图 4 小椭圆油罐示意图

§ 7.4 程序

程序 youliang1

```
load jinyoul.txt
```

```
load chuyoul.txt
```

```
g1=jinyoul(:,2)*10^(-2);
```

```
g2=chuyoul(:,2)*10^(-2);
```

```
syms a b h l
```

```
a=0.89*10;b=0.6*10;l=2.45*10;
```

```
V=(a*l/b).*((h-b).*((2.*h.*b-h.^2).^0.5)+b.^2.*asin((h-b)./b)+0.5.*b.^2*pi);
```

```
V1=subs(V,h,g1)
```

```

V2=subs(V,h,g2)
h1=jinyou1(:,1)+262;
h2=3706.91+262-chuyou1(:,1);
c1=V1-h1
c2=V2-h2
x1=abs(V1-h1)./V1
x2=abs(V2-h2)./V2
plot(g1,V1,'.r',g1,h1,'.c')
xlabel('进油后储油量');
ylabel('罐内油位高度');
title('罐内油位高度随进油后储油量的变化情况')
legend('储油理论量','储油实际量')
grid on
figure(2)
plot(g2,V2,'.r',g2,h2,'.c')
xlabel('出油后储油量');
ylabel('罐内油位高度');
title('罐内油位高度随出油后储油量的变化情况')
legend('储油理论量','储油实际量')
grid on
figure(3)
subplot(2,1,1)
plot(g1,c1,'.b')
xlabel('罐内油位高度');
ylabel('进油后储油量的误差');
title('进油后储油量的误差随罐内油位高度的变化情况')
grid on
subplot(2,1,2)
plot(g2,c2,'.b')
xlabel('罐内油位高度');

```

```

ylabel('出油后储油量的误差');
title('出油后储油量的误差随罐内油位高度的变化情况')
grid on
程序 youliang2
load jinyou2.txt
load chuyou2.txt
g1=jinyou2(:,2)*10^(-2);
g2=chuyou2(:,2)*10^(-2);
syms a b h m al
a=0.89*10;
b=0.6*10;
a1=0.4*10;
L=2.45*10;
m=4.1*pi/180;
k=(a1*tan(m)/b)-1;
p=(h+a1*tan(m)-b)/b;
q=(h+a1*tan(m)-b-L*tan(m))/b;
V=-((-1/3)*(1-q.^2).^(3/2)+(1-q.^2).^(1/2)+q.*asin(q)+1/3.*(1-p.^2).^(3/2)-(1-p.^2).^(1/2)-p.*asin(p)+pi*(q-p)*0.5).*a*b^2./tan(m)
V1=subs(V,h,g1)
V2=subs(V,h,g2)
h1=jinyou2(:,1)+215;
h2=3299.74+215-chuyou2(:,1);
c1=V1-h1
c2=V2-h2
x1=abs(V1-h1)./V1
x2=abs(V2-h2)./V2
plot(g1,V1,'.r',g1,h1,'.c')
xlabel('倾斜时进油后储油量');
ylabel('罐内油位高度');

```

```

title(' 倾斜时罐内油位高度随进油后储油量的变化情况')
legend(' 储油理论量', ' 储油实际量')
grid on
figure(2)
plot(g2, V2, '.r', g2, h2, '.c')
xlabel(' 倾斜时出油后储油量');
ylabel(' 罐内油位高度');
title(' 倾斜时罐内油位高度随出油后储油量的变化情况')
legend(' 储油理论量', ' 储油实际量')
grid on
figure(3)
subplot(2, 1, 1)
plot(g1, c1, '.b')
xlabel(' 罐内油位高度');
ylabel(' 倾斜时进油后储油量的误差');
title(' 倾斜时进油后储油量的误差随罐内油位高度的变化情况')
grid on
subplot(2, 1, 2)
plot(g2, c2, '.b')
xlabel(' 罐内油位高度');
ylabel(' 倾斜时出油后储油量的误差');
title(' 倾斜时出油后储油量的误差随罐内油位高度的变化情况')
grid on

```